

AI CHATBOT FOR FINANCIAL CRIME ANALYSIS

HAYDEN HUAN KEE JIUN

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AI CHATBOT FOR FINANCIAL CRIME ANALYSIS

HAYDEN HUAN KEE JIUN

A thesis submitted in fulfilment of the
requirements for the award of the degree of
Bachelor of Computer Science (Computer Networks & Security)

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I hereby declare that I have read this thesis and in my opinion this thesis is sufficient in term of scope and quality for the award of the degree of Bachelor of Computer Science (Computer Networks & Security)

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In preparing this thesis, I was in contact with many people, researchers, academicians, and practitioners. They have contributed towards my understanding and thoughts. I wish to express my sincere appreciation to my main thesis supervisor, TS. DR MOHD ZAMRI BIN OSMAN, for encouragement, guidance, critics and friendship. Without his continued support and interest, this thesis would not have been the same as presented here.

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ABSTRACT

The criminally complex and continuously growing numbers of financial crimes such as money laundering, fraud and financing of terrorism are major challenges to the Malaysian financial investigators. Even with sound legislations like the Anti-Money Laundering, Anti-Terrorism Financing and Proceeds of Unlawful Activities Act 2001 (AMLA) and effective regulation of Bank Negara Malaysia (BNM), recent manual investigation procedures are slow, may have incorrect human decision substantiated errors, and cannot keep abreast with the manner in which the illicit activities are progressing, which sometimes result into possible misinterpretation and missing important decision casting. This makes the procedures even more complicated when many agencies are involved, investigators should follow various guidelines and legal documents. To cover these crucial operational mistakes, FinGuardMY is under development as a unique AI chatbot dedicated to the financial crime investigation unit in Malaysia to provide its investigators with the ability to automate the process of analyzing financial operations and the guarantee to constantly check their operations against the law. Based on the powerful application of Natural Language Processing (NLP), LangChain, Flowise AI, and LLaMA large language model, FinGuardMY will allow users to create natural language requests, simply import their documents, and get an in-depth legal explanation, based on Malaysian laws. The architecture of the system is based on the FastAPI backend and React.js for frontend and assisted by MySQL as the source of structured information and Pinecone to store and retrieve semantic data quickly. Most importantly, FinGuardMY incorporates OWASP security to protect confidential financial data and security of the API and LLM with the possibility to prevent such vulnerabilities, as prompt injection and release of sensitive information. Finally, FinGuardMY stands the chance of making the investigation process very simple, highly accurate, reduce the risk of human errors, and offers a localized, enlarging solution that perfectly aligns with the Malaysian special regulatory environment. Through adoption of this project, Malaysia can work towards the development of smarter and more secure digital tools to tackle effectively financial crime and improve financial stability and integrity of the nation.

ABSTRAK

Peningkatan dalam kompleksiti dan kuantiti jenayah kewangan seperti pengubahan wang haram, penipuan dan pembiayaan keganasan memberikan cabaran besar kepada agensi penguatkuasaan kewangan di Malaysia. Meskipun wujud perundangan tegas seperti Akta Pencegahan Pengubahan Wang Haram dan kawal selia oleh Bank Negara Malaysia, kaedah penyiasatan secara manual yang masih digunakan hari ini memerlukan masa yang panjang, mudah terdedah kepada garis kesilapan manusia, dan lemah dalam menghadapi taktik jenayah yang semakin canggih. Penyiasat biasanya mesti memeriksa kebanyakan rekod transaksi dan dokumen undang-undang dengan tangan, yang boleh menyebabkan kesilapan dalam tafsiran dan melambatkan proses penyiasatan kritikal. Projek ini memperkenalkan FinGuardMY, sebuah AI Chatbot yang direka khas untuk membantu penyiasat di Malaysia menganalisis kes jenayah kewangan dan menilai pematuhan terhadap undang-undang secara automatik dan dalam masa nyata. Untuk membuat AI Chatbot ini teknologi canggih seperti LLM, LLaMa 2, LangChain dan Flowise AI digunakan untuk memudahkan pengguna untuk berinteraksi melalui pertanyaan dalam bahasa normal, muat naik dokumen mereka dan mendapat penjelasan undang-undang yang relevan. Bahagian belakang sistem dibina menggunakan FastAPI manakala antara muka pengguna dibina menggunakan React.js manakala sistem pangkalan data menggabungkan MySQL dan Pinecone. Sistem ini juga mematuhi garis panduan keselamatan OWASP bagi API dan LLM bagi mengurangkan risiko kerentanan seperti serangan Prompt Injection dan pendedahan maklumat sensitif. Di samping itu, AI Chatbot ini bukan sahaja mempercepatkan proses penyiasatan, malah mengurangkan risiko kesilapan manusia, menyediakan penyelesaian yang berskala dan selaras dengan konteks penguatkuasaan undang-undang kewangan di Malaysia. Projek ini mendorong transformasi digital ke arah sistem lebih cerdas dan selamat dalam memerangi jenayah kewangan di negara ini.

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LIST OF ABBREVIATIONS

NFCC	-	National Anti-Financial Crime Centre
RAG	-	Retrieval-Augmented Generation
API	-	Application Programming Interface
AI	-	Artificial Intelligence
OWASP	-	Open Worldwide Application Security Project
BNM	-	Bank Negara Malaysia
AMLA	-	Anti-Money Laundering, Anti-Terrorism Financing and Proceeds of Unlawful Activities Act 2001
LLM	-	Large language models
UAT	-	User Acceptance Testing
UML	-	Unified Modelling Language
ERD	-	Entity Relationship Diagram

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CHAPTER 1

INTRODUCTION

1.1 Introduction

In today's digital world, the financial sector is growing faster than ever. However, this rapid growth also brings serious challenges, especially crimes like money laundering, fraud, and terrorism financing. Even though Malaysia already has strong laws like the Anti-Money Laundering Act (AMLA) and follows guidelines from Bank Negara Malaysia (BNM), spotting these crimes as they happen is still very hard.

Hence, whenever this occurs, investigators usually find themselves to go through massive quantities of data manually currently. This is time consuming and may result to errors or causes delays. At the same time, there are generally numerous numbers of agencies that are involved in cases of financial crimes, making the situation further complex. Currently, Malaysia is in the process of digital banking as a blueprint in the Financial Sector 2022-2026. Due to it, wiser and quicker tools that are to be used to combat financial crime are critically needed.

The concept of this project is using AI chatbot to analyse the financial crime cases and generate suggestions to assist investigators. This chatbot will apply the use of Natural Language Processing (NLP) and machine learning to scan the document, identify suspicious patterns and analyse against Malaysia financial laws. Then it will provide the recommendation to investigators.

1.2 Problem Background

Malaysia's expanding financial sector is facing with ongoing issues like money laundering, terror financing, and fraud. The financial institutions are still unable to track sophisticated illicit activities in real time even under stringent regulations like AMLA and even with oversight by BNM.

The issue here is that investigators must go through large amounts of documents on an individual basis. That the manual procedures are time-consuming is often a source of error. As an example, investigators may misinterpret some essential information because they are exposed to quite large or complicated datasets. A very good example of such an issue is the scandal of 1MDB. In that instance, vast figures were shifted from one account to another.

Another problem is the presence of numerous agencies responsible for fighting cases of financial crime. Typically, the agencies comprise law enforcement such as police, Bank Negara Malaysia (BNM), and the Securities Commission and global groups. That is why investigators normally must contend with a variety of guidelines, paperwork, and legal work documents from different agencies. In addition to that, legal and compliance regulations are subject to change. As such, investigators are required to keep abreast of the latest legal and regulatory developments.

As a result, the use of AI Chatbot that can understand legal laws and can analyse financial data may significantly decrease the frequency of human errors. Simplifying the inquiry and presenting the insights to it facilitate investigators in synthesising their decision-making. In the long term, it becomes a valuable resource in terms of promoting Malaysia's financial stability and the provision of justice services.

1.3 Project Aim

The aim of this project is to develop an AI chatbot that assists investigators in analysing financial crime cases by automating legal compliance checks and document analysis.

1.4 Project Objectives

The objectives of the project are:

- (a) To study the patterns in financial transaction data that show possible breaches of law under Malaysian financial laws.
- (b) To design and develop a secure, RAG-based AI chatbot that helps investigators analyse financial crimes.
- (c) To validate and test the security measures and the effectiveness of the AI chatbot aligned with OWASP standards.

1.5 Project Scope

The scopes of the project are:

- (a) The AI chatbot focuses only on Malaysian financial crime laws and does not include international or non-financial crime regulations.
- (b) The security features of the AI chatbot follow OWASP standards.
- (c) The AI chatbot can provide a clear explanation of detected irregularities and potential legal violations.

1.6 Project Importance

Creating such an AI chatbot is imperative as cases of financial crimes in Malaysia are on the rise and are now more complicated to deal with. In addition to impacting economic stability, such crimes erode public confidence in the system. As the issue widens out, it becomes normal to understand how much the sophisticated and efficient tools to assist the investigators are required. The goal targeted by this project is to implement an RAG-based AI chatbot capable of handling financial information and minimising the errors caused by humans and giving legal counsel compatible with Malaysian law. Unlike readily available technology, prone to becoming overly complicated or even lacking a local uniqueness, this chatbot targets accessibility, security, and localisation to cater to the requirements of a particular area. The process of investigations could be streamlined by using this chatbot, the status of decision-making could be boosted, and the burden on the enforcement officers could be alleviated.

1.7 Report Organization

This is a thesis on the development of FinGuardMY AI chatbot that helps solve intricate crimes in the financial sector in Malaysia. Chapter 1 presents the background, objectives and the relevancy of the project. Chapter 2 compares the current approaches to financial crime detection and identifies the inadequacies of those solutions as well as the novel implementation of FinGuardMY under the law of Malaysia. Chapter 3 explains how to work with the Agile method, including the stages of collection of requirements, development, implementation, testing, and deployment with such libraries as React.js, FastAPI, MySQL, Pinecone, LangChain, and Flowise AI. The chapter of Requirement Analysis and Design introduces the requirements of the field, UML diagrams, ERD used to design a database, and the UI prototypes developed with the help of Figma. Conclusion lists what has been done at the PSM1 phase, ensures the initial objectives are met, makes an approximate summary of the initial test results, and draws a plan to improve AI capabilities, as well as testing and deployment during PSM2.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

In this chapter, the literature, current systems, and technologies in developing an AI chatbot that can investigate financial crime in Malaysia and other countries are reviewed. With the sophistication level of the financial crimes like money laundering, fraud and corruption, they are a major challenge to the government and law enforcement agencies in the country. Modern financial schemes are too complex to be manually discovered in time and hence a technological method is necessarily needed to help investigators.

An effective solution would be provided by this project. FinGuardMY, Retrieval-Augmented Generation (RAG) AI-based chatbots can become the tool to empower investigators by giving them direct access to legal statutes and case precedence in a real-time manner. Although major agencies such as Bank Negara Malaysia (BNM) started implementing AI, there is still an important limitation since most of the international systems are poorly integrated with the Malaysian local system due to the non-compatibility between the Malaysian criminal law and most of the international systems, such as the Anti-Money Laundering, Anti-Terrorism Financing and Proceeds of Unlawful Activities Act 2001 (AMLA).

Thus, this review considers the existing environment in a systematic way and forms the basis of the targeted solution. It compares the current tools existing in the fields of AI in finance and law, explores the essential technologies such as Natural Language Processing (NLP) and machine learning, and discusses which current tools best serve the needs of the architecture and framework used to create the tool that best addresses the individual needs of financial crime investigators in Malaysia

Figure 2.1 shows the organizational chart of the NFCC. On top lies the Director General who is the head of the entire organization. The Deputy Director General and four Directorate divisions, namely, Integrated Operation Division, Strategic Research Division, Asset Management Division, Legal Division, Management Services Division and, Internal Audit and Integrity Unit, report to the Director General. Hierarchical nature is visually displayed in the chart, whereby the first three divisions are controlled by the Deputy Director General, whereas the remaining three divisions report to the Director General.

Organizational Chart

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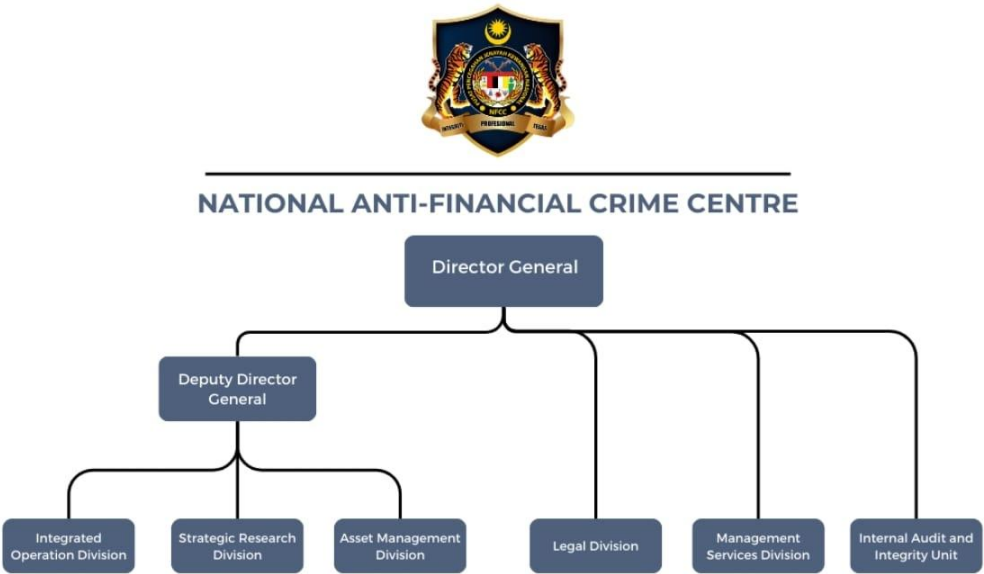


Figure 2.1 Organizational Chart for National Anti-Financial Crime Centre

2.2 Project Data Source

For FinGuardMY AI Chatbot to generate and provide accurate and useful legal information, this project uses elaw.my as the main data source to get the legal knowledge for the chatbot.

2.2.1 Overview of elaw.my

eLaw.my is the biggest and largest online Malaysian law database which has more than 100,000 court judgments of different Malaysian courts such as Federal or Supreme Court, Court of Appeal, High Court, Industrial Court, and Syariah Court and back dated to the 1900s (About eLaw, 2025). It also offers up-to-date Federal and State legislation and municipal by-laws and amendments in the form of a timeline that includes annotations and cross-references between Annotations and current cases. The platform includes an effective and simple search engine, enabling the legal professional to effectively search for cases and legislation in cross-referencing, viewing case citations in the local legal journals, and precedent maps charting the relationship between cases or represented in a list of cases are said to be considered, followed, overruled, or distinguished. This renders this eLaw.my as the key resource in the legal research and practice in Malaysia, that is intended to assist busy people in the Malaysian legal community with the multi-functional search features and extensive legit legal materials (About eLaw, 2025).

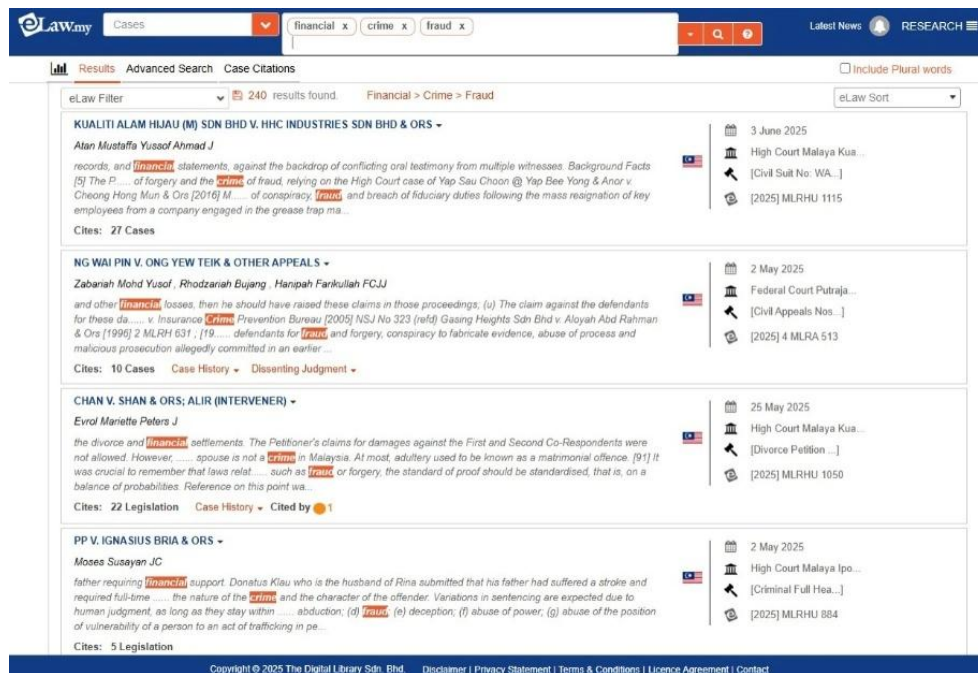


Figure 2.2 Platform for elaw.my

2.2.2 Data Scope

The data scope of this project is focused on financial crimes such as money laundering. The elaw.my platform will be the only provider of all the documents that will be employed in training the RAG-based version of the chatbot. This data will be assessed within a one-year period. There are 2 examples of financial crime cases get from elaw.my as dataset for this project:

(a) ROZIAH HARUN v. INSTITUT INTEGRITI MALAYSIA



Figure 2.3 Sample Case 1

(b) NORIHAN TALIB lwn. PENDAKWA RAYA

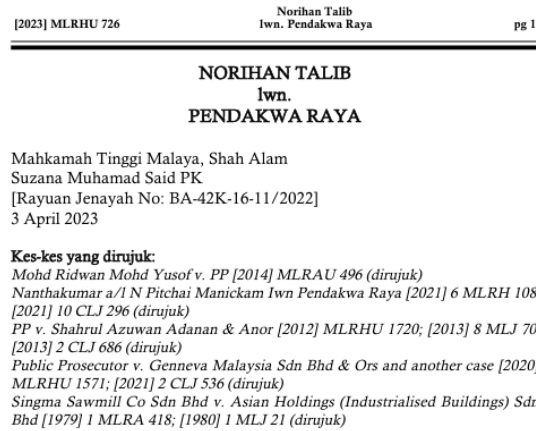


Figure 2.4 Sample Case 2

2.3 Current System Analysis

In the current system, investigators still depend heavily on manual work to handle financial crime cases. They often must go through huge amounts of transaction records, then compare them with various legal documents line by line, trying to spot anything suspicious and this is slow and exhausted. In most cases, they simply run on simple tools such as spreadsheets or simple graphic charts that are not smart enough to identify more deeper and more complex indications towards the financial crime. This way, critical hints could be overlooked very quickly, and investigations could last much longer than desired.

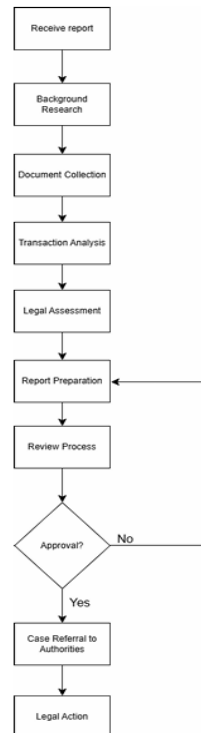


Figure 2.5 Current System Workflow

To accomplish the task, the current system of financial crime investigation in National Anti-Financial Crime Centre (NFCC), depends mostly on manual procedures. The investigators have the heavy duty to conduct thorough audit of vast number of financial records to determine and confirm whether there is adherence as per the relevant acts such as the Anti-Money laundering, Anti-Terrorism Financing and Proceeds of Unlawful Activities Act 2001 (AMLA). The largely manual methods are inefficient because of their inability to face the constantly evolving methods that criminals use when committing financial crime. This will also cause human-error and time-consuming when manually analyse the documents.

For example, a financial institution files a Suspicious Activity Report (SAR) reflecting a group of high dollar volume transactions that were purposefully arranged to just under-reporting requirements imposed by the AMLA laws. In the present investigative process, there is usually a time lapse of an inter-agency data gathering that can take one to two weeks. Then, the manual processing of each such volume of transactional information may extend the time horizon a particular investigation to

over a month, and this aspect alone presents a high risk of the investigators missing essential interconnection, including linkages to known shell companies on watch lists.

Moreover, there is an absence of immediate and combined legal assistance in perfecting the confronting and implementation of complex AMLA and Bank Negara Malaysia (BNM) laws. Another limitation is added to the report generation process, which prolongs the process of investigations and may give illegal participants enough time to launder illegal funds. Such operational problems present long-term difficulties that may affect the organization in terms of inefficiency and lower the morale of the National Anti-Financial Crime Centre (NFCC) staff. The limitation of this manual process further highlights the strength of the necessity to introduce a more efficient and automated process that could enhance the capabilities of the federal and state agencies to address financial crimes in Malaysia.

2.4 Comparison between existing systems

This section discusses the comparison between existing systems of law firm AI chatbot which are Clio Duo, Legora, Harvey AI with the proposed solution, FinGuardMY AI Chatbot.

2.4.1 Clio Duo

Clio Duo is an AI legal assistant installed back into Clio practice management software, which, used by law firms. It is supposed to automate the routine activities such as document draughting, appointments, and communication to clients. Lawyers can interact with information on their cases with natural language input instead of searching calendars or files manually. The assistant provides access to solid information in a real-time manner, increases automation of activities, and optimises time. Even though Clio Duo is most optimal when requiring more productivity of law practices, the relying on Clio ecosystem prevents its further adaptability to the general

legal practice and more extensive functions associated with compliance and investigation measures (Bennett, 2025).

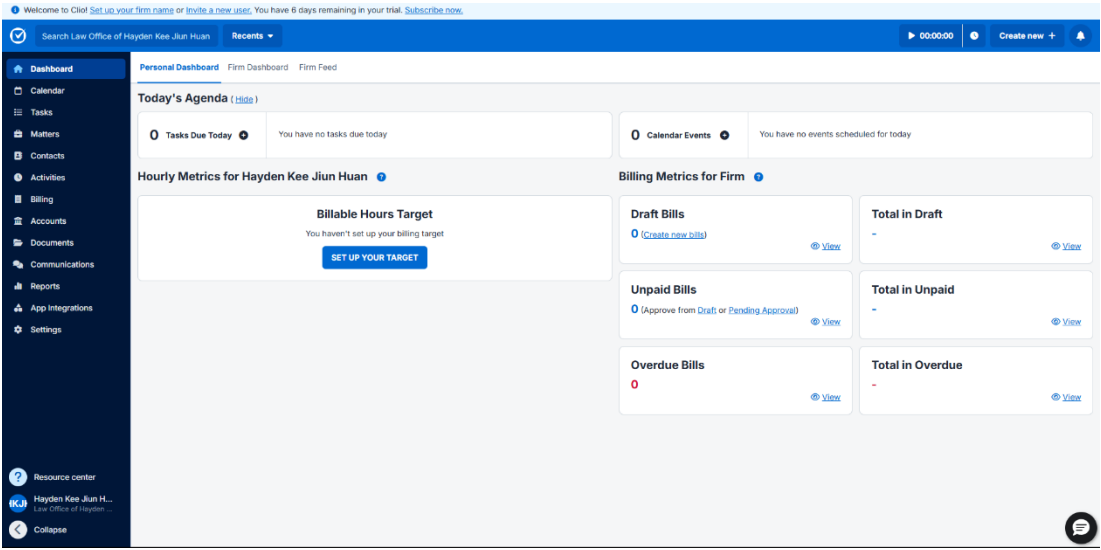


Figure 2.6 Clio Duo User Interface

2.4.2 Legora

Legora is an AI system that automates the document process and legal contract analysis. It helps the legal professionals find the most crucial terms, identify potentially risky provisions, and give certainty that the document corresponds to the typical legal and regulated templates. Legora uses a mixture of machine learning and rule-based systems to identify faster review processes and lower rates of manual error draughting. It supports lifecycle management in contracts and thus it can be used by the in-house law teams, as well as law firms (LEGORA | Collaborative AI for Lawyers, n.d.).

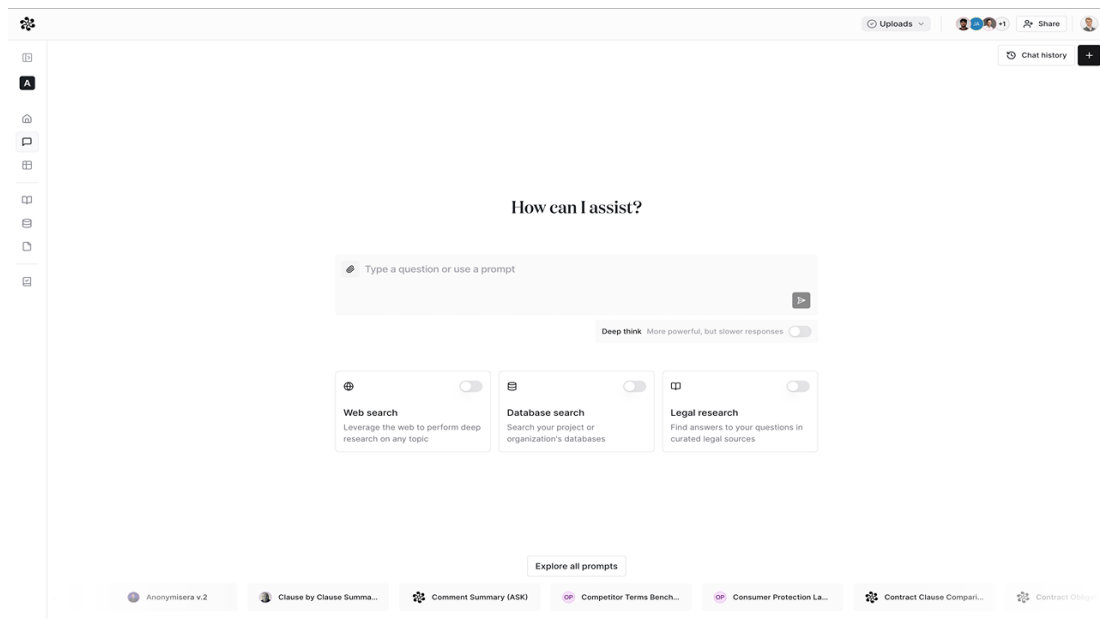


Figure 2.7 Legora User Interface

2.4.3 Harvey AI

Harvey AI is a high-end legal software created with the assistance of leading international law firms and running on OpenAI's large-scale language models. It offers document analysis, case strategy assistance, contract draughting, and draughting of legal opinions in different jurisdictions. It can integrate with firm knowledge bases and processing complex legal questions effectively. Harvey is capable and scalable but lacks the depth of legal language of some of the more specialised tools. It is optimised for high-end legal environments and therefore more appropriate for large firms but might need extensive customisation for specific domains such as conducting an anti-money laundering investigation in a Malaysian scenario (Rattray, 2025).

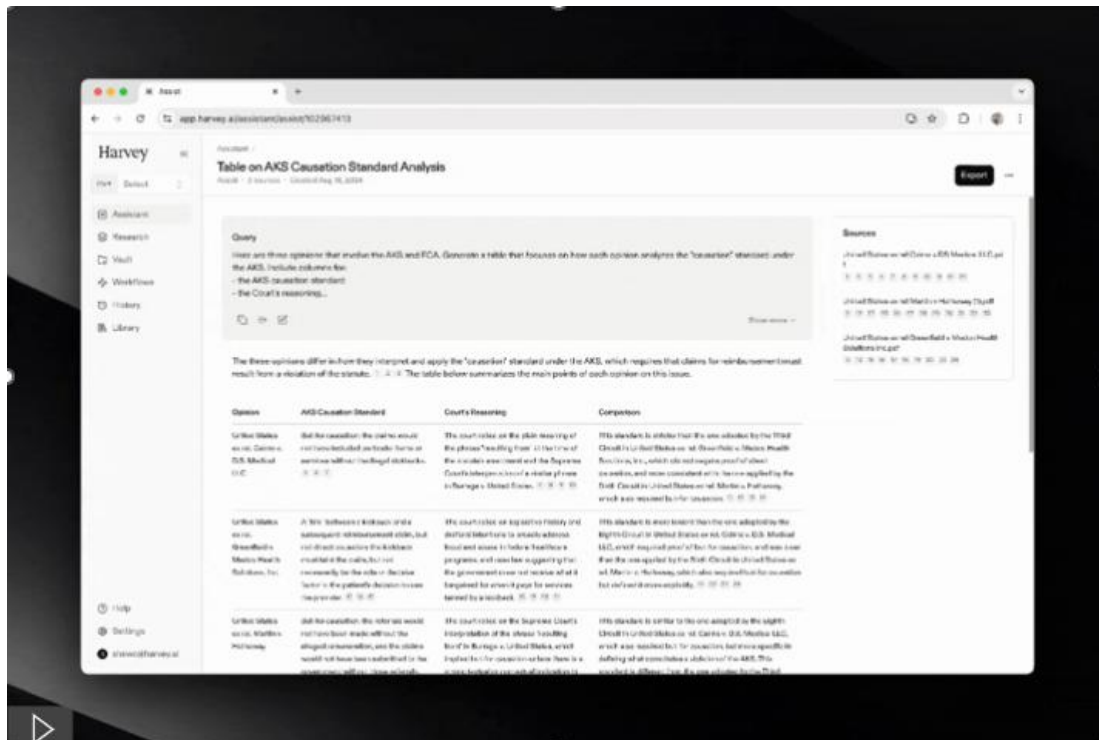


Figure 2.8 Harvey AI User Interface

2.4.4 Proposed Solution - FinGuardMY

With a focus on improved legal compliance, FinGuardMY aims to provide a tailor-made AI chatbot to the Malaysian financial institutions to pursue financial crimes and keep up to date with local legislations. With the integration of NLP and ML, the chatbot can give answers regarding laws that relate to the cases. The integration of the features of LangChain and Flowise AI enables FinGuardMY to deliver a user with more informed and personalised interactions addressing such areas of concern as legal information, and compliance training.

Table 2.1 Comparison Between Existing System

Feature/System	FinGuardMY	Clio Duo	Legora	Harvey AI
Primary Function	Financial crime detection and compliance guidance	Legal assistant for law firm task management	Contract analysis and legal document automation	Advanced legal analysis and document drafting

	tailored to Malaysian laws			
User Interaction	Text-based conversational interface	Text-based assistant integrated with Clio	Web interface with AI contract workflows	Text-based with integration into legal workflows
Regulatory Focus	Malaysian AMLA, SC guidelines, BNM policies	General law practice management	Contractual compliance, general legal document review	Multi- jurisdictional legal systems
Technology Stack	NLP, ML, LangChain, Flowise AI	AI assistant with Clio's cloud ecosystem	NLP, ML, custom rule- based logic	OpenAI LLMs integrated with firm knowledge bases
Deployment Model	Web-based chatbot	SaaS platform with Clio	SaaS platform	Cloud-based enterprise solution
Adaptability	High	Moderate	Moderate	High

2.5 Literature Review of Technology Used

This section discusses the technology used in this project which are React and Tailwind CSS for frontend, FastAPI for backend, MySQL and Pinecone serve as databases and for Large Language Model (LLM) technologies like Langchain, Ollama and Flowise AI will be used.

2.5.1 Frontend Technology

- (a) React: An easy-to-use JavaScript library to construct user interfaces, and especially single-page applications. It was designed by Facebook and launched on the 29th of May 2013. React enables developers to build scalable UI and update the DOM with the idea of a virtual DOM rapidly (Abramov & Nabors, 2023).



Figure 2.9 React

- (b) Tailwind CSS: Utility-first CSS framework enabling the construction of custom user interfaces, without the need to ever leave the HTML. It is owned by Adam Wathan and Stephen Schoger, and it was launched in November 2017. Tailwind CSS focuses on the responsive and very customizable design system (Budiman, 2023).



Figure 2.10 Tailwind CSS

2.5.2 Backend Technology

FastAPI: A fast, modern, web, framework to create Python 3.6+ APIs using standard Python type hints. It was authored by Sebastián Ramírez and published in December 2018 (Technocrat, 2023).



Figure 2.11 FastAPI

2.5.3 Databases

- (a) Pinecone: A kind of vector database, where similarity search can be performed quickly. Pinecone Systems released Pinecone in 2020. It supports high-dimensional data and is optimised to run in real time in use cases such as recommendation systems and natural language processing (Garg, 2023).



Figure 2.12 Pinecone Vector Database

- (a) MySQL: A relational database management system (RDBMS) that is open-source and is utilized in storing and retrieving structured data. Firstly, it was published on May 23, 1995, by MySQL AB (presently possessed by Oracle Corporation) and because of its stable, solid performance, it is among the most circulated databases (Erickson, 2024).



Figure 2.13 MySQL Database

2.5.4 Large Language Model (LLM)

- (a) Langchain: A recipe to build applications using language models and supporting external data and context in memory. It is proposed by Harrison Chase in October 2022 and improves the functions of the large language model on activities such as question answering and chatbots (TechLatest.Net, 2024).

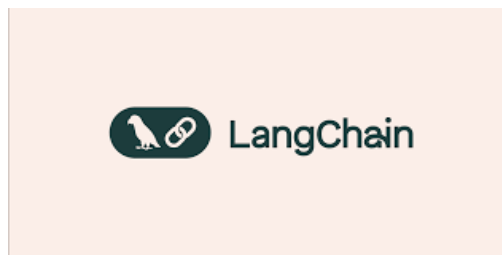


Figure 2.14 LangChain

- (a) Ollama: A locally hosted open-source framework to use large language models. Developed in 2023 by the Ollama team, it enables users to deliver and tune the LLMs without depending on cloud services, with the concentrated centralization on privacy and advantage (Tahir, 2025).



Figure 2.15 Ollama

- (a) Flowise AI: A low-code, open-source tool that allows creating custom applications and workflows using LLM. It is a simplified version of creating AI solutions and will be introduced by the Flowise team in 2023, having made the process of creating such solutions available even to non-experts (Tony, 2024).



Figure 2.16 Flowise AI

2.5.5 Security

- (a) OWASP: A nonprofit organization that offers resources and tools that can be used to enhance software security, such as the OWASP Top Ten list of vulnerabilities. It was established in September 2001 and has become a world community whose focus is the improvement of web application security (Barracuda, 2024).
- (b) OWASP Top 10 API Security Risks: It outlines the most potential vulnerabilities in Application Programming Interfaces (APIs). It discusses API security risks such as unauthorized access to data due to poor implementation of object-level authorization, user impersonation through improper authentication, data exposure, denial-of-service attacks because of insufficient rate limiting, unauthorized access to functions, unintended summons of the server, unsecure settings, injection-based threats that trigger the breach of data, unmonitored API routes, and the dangers of unvalidated third-party APIs.
- (c) OWASP Top 10 for Large Language Model Applications: This is fundamental security and ethical risks of the systems based on large

language models. These are steering with malicious inputs, accidentally exposing sensitive information, poisoning models with corrupted training data, overwhelming systems with excessive query resources, insecure dependencies, insecurely written plugins, unintended autonomous behaviour, data access limitations with vector databases, incorrect model outputs due to unverified output, and theft of proprietary models.



Figure 2.17 OWASP

- (d) Postman: A collaboration platform for API development, used for testing, documenting, and monitoring APIs, with security testing features. Initially released in 2012 by Abhinav Asthana, Ankit Sobti, and Abhijit Kane, it has evolved into a comprehensive tool for API lifecycle management (Rodrigues, 2024).



Figure 2.18 Postman

2.6 Chapter Summary

The chapter provides an overview of the key literature and different systems implemented for creating an AI chatbot for financial crime investigation in Malaysia. Additionally, the chapter presents the key weaknesses of Malaysia's manual investigation of financial crime, which include wide data handling. The chapter compares the Clio Duo, Legora Harvey AI, and the introduced FinGuardMY with the help of their pros and cons. There is a distinct interest in the lack of proper compliance consideration in global Malaysian solutions. Also, the chapter investigates the implementation of the project technologies, like NLP, LangChain, and Flowise AI, in explaining how these tools are conducive to transaction processes' automation and the provision of regulatory measures. The justification of the comparison provides a solid basis for the design and development of the targeted AI chatbot to serve the practical requirements of the Malaysian financial crime investigators in terms of efficiency, effectiveness, and protection.

CHAPTER 3

SYSTEM DEVELOPMENT METHODOLOGY

3.1 Introduction

In this chapter, the method applied to develop the AI chatbot for financial crime analysis is presented. It is the roadmap of the technical and analysis process of the project. It also describes the methodology applied to this project in a clear way, the process of each phase of the applied methodology, the technology or tool applied to develop this system, and the system requirement analysis. The aim of this chapter is to achieve a systematic, effective, and efficient development process that matches the project goals to produce an effective solution.

3.2 Methodology Choice and Justification

This section will discuss the methodology choice and justification for this project:

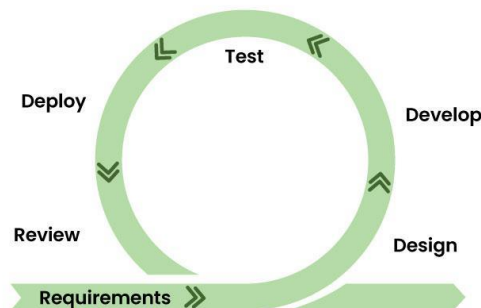


Figure 3.1 Agile Methodology (GeeksforGeeks, 2024)

Agile methodology is chosen to manage this project as it is iterative with a feedback focus and is very suitable in the case of developing FinGuardMY. Such a strategy will additionally help in making the system further the requirements of the real work that is a key stakeholder who is the National Anti-Financial Crime Center (NFCC) Lab with whom we collaborate and receive feedback continually. The flexibility Agile inherently offers is essential in dealing with technicalities around the project, such as integrating Natural Language Processing (NLP) and Flowise AI, as well as remaining adaptive in terms of changing legal and compliance considerations. Essentially, Agile offers the reactive and dynamic framework that is imperative in maneuvering the technical and regulatory environment of financial crime analysis in Malaysia.

3.3 Phases of the Chosen Methodology

This section will outline the activities and operation of FinGuardMY AI Chatbot in all the phases of Agile methodology.

3.3.1 Requirement Gathering

During the first step, requirement gathering was carried out to determine the scope of the FinGuardMY AI Chatbot. In discussions with academic supervisor, who served as a proxy representing the National Anti-Financial Crime Centre (NFCC) Lab, some important functional requirements, those being the analysis of documents, the generation of legal recommendations and the retrieval of a history of chats will be mentioned and discussed. Such key non-functional requirements as the system security, scalability, and adherence to the Malaysian financial regulations were also formulated. All the collected requirements were then prioritized and put in a backlog as a product backlog, which was refined with the repeated feedback of stakeholders to keep it in line with the requirement of the project and operational requirements of financial crime investigators.

3.3.2 Design the Requirements

In the design phase, the collected requirements were converted into architectural and user interface diagrams. The Unified Modelling Language (UML) diagrams were created in draw.io. At the same time, it was developing prototypes of UI/UX in Figma, including login page, chat interface of an investigator, as well as administrative dashboards. These graphic designs were shown to the stakeholders to receive a response and further refined. Sprint planning was done during this phase where designs were analysed using functional requirements and heuristics of using the product to provide a strong development platform that was user oriented.

3.3.3 Construction

FinGuardMY AI Chatbot will be developed in stages and phases based on bi-weekly iterations. It was implemented with a modern technology stacks, the frontend was using a React framework and Tailwind CSS to provide dynamic user interface, backend was using FastAPI framework to handle API calls and business logic, the MySQL database was used to save structured data about users and cases, and Pinecone cloud was used to manage vector embeddings which are necessary to conduct semantic analysis by the AI. Every sprint ended with a functioning and integrated increment of the system that was reviewed at once thus providing a technical fluidity and permitting an early identification of problems.

3.3.4 Functional & Non-Functional Testing

To ensure the reliability, security, and performance, the testing process was a part of the product that took place throughout its existence. Integration testing was used to provide smooth communication among the React.js frontend and FastAPI backend, AI models, and databases. The area of security testing was of primary importance, and the mitigation of such aspects as SQL injection and prompt injection should be based on the OWASP directions. Lastly, we performed a User Acceptance

Testing (UAT) with the National Anti-Financial Crime Centre (NFCC) Lab to determine that the system fulfilled their usability and functional requirements, and their responses directly became the input into new development sessions.

3.3.5 Deployment

At the deployment stage, the FinGuardMY AI chatbot was transferred to a stable and safe environment of production. The system was implemented in a cloud-based server it was provided with the required resources that can provide performance and scaling. An anonymized sample of data on financial crimes was used to pilot test and therefore, eliminate performance bottlenecks before a complete roll out. The last setting was the proliferation of all APIs, databases, and AI models and the hardening of the security by establishing the HTTPS and Role-Based Access Control (RBAC).

3.3.6 Feedback

The continuous improvement was supported by systematic feedback that is inherent in the Agile framework. A Sprint Review was a meeting that took place at the end of every sprint according to which the recent system increment should be demonstrated to such National Anti-Financial Crime Centre (NFCC) Lab. Their suggestions regarding functionality, usability, and conformity to the legal standards were recorded and incorporated into the mechanism of setting priorities in the product backlog during the upcoming sprint, for example, making the legal advice generated by AI further user-friendly was deemed as one of the crucial aspects based on their suggestion.

3.3.7 Design Modelling

With Unified Modelling Language (UML) diagrams allowed system design to facilitate clarity and uniformity in development. The use case diagram established the connection between the system and the investigators in such scenarios as upload documents and analyse document, and view chat history. The activity diagram represented the workflow for investigators to use this system.

Class diagram presented the internal structure of the system, including major classes like Users, Case, ChatSession and Subject, and their attributes and methods. Sequence diagram illustrates the progress of investigators using the system. The models of design confirmed that functionalities were in accordance with the system behaviour and requirements.

3.3.8 Design Tools

The design process was smoothed out by two major tools. Draw.io was used to design use case, activity, class, and sequence diagrams. Its web-based, user-friendly interface allowed rapid and flexible modelling of fragments of system and flows of work. They played a key role in the correct translation of the stake requirements into technical structure.

Designing User interface was done by using Figma. It provided an environment of visual prototyping to experiment with UI of the chatbot and make it easy to make use of. Using Figma, the user flow and layout could be iterated on in real time to increase its readability and accessibility by the potential users such as financial crime investigators.

3.4 Tools and Technology Used for System Development

This section will discuss about the tools and technology used for FinGuardMY AI Chatbot:

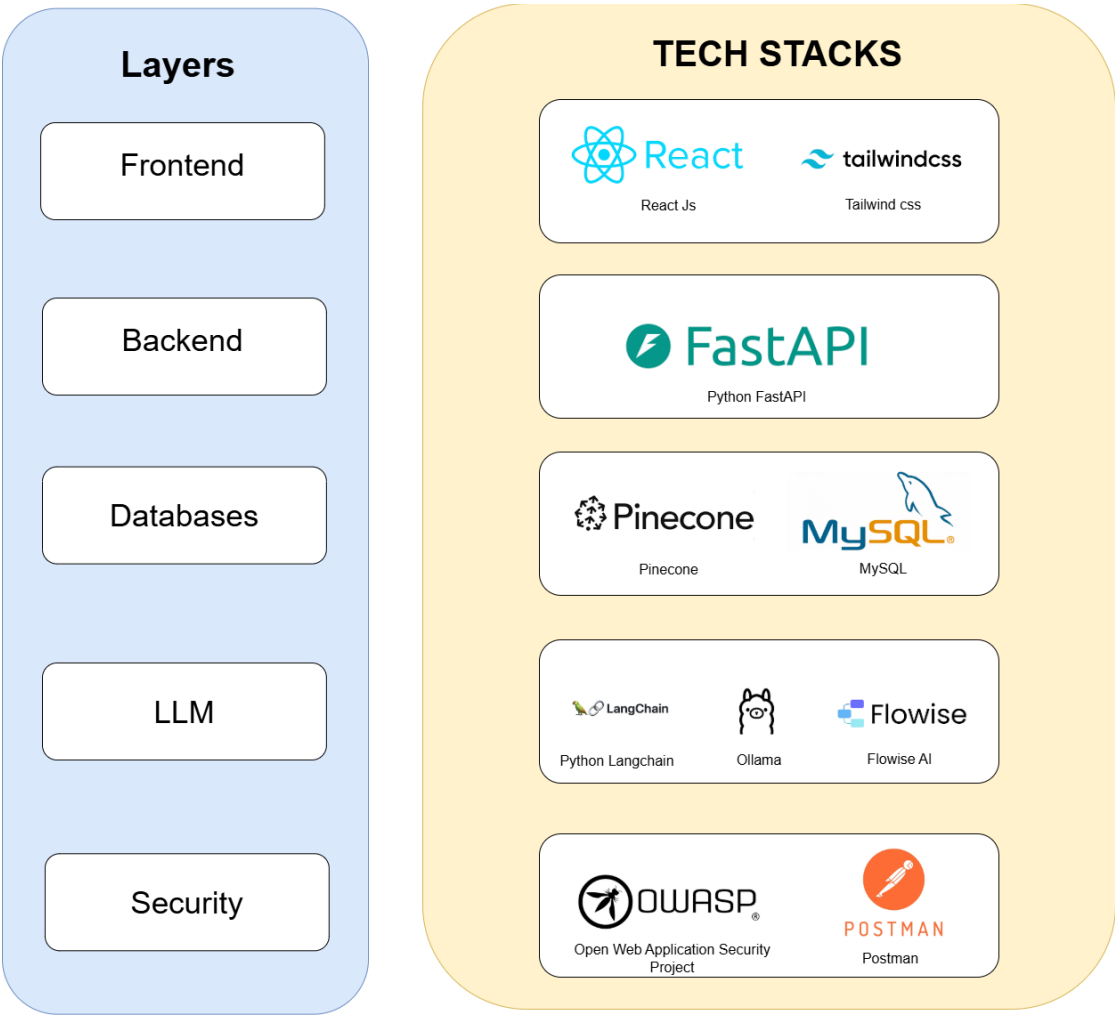


Figure 3.2 Tools and Technology used

3.4.1 Frontend Development

React is used as the main frontend technology in FinGuardMY because it allows for the construction of responsive and dynamic interface for a website with its ‘hooks’ and rendering feature (Abramov & Nabors, 2023). By using separate parts of the system, the features like chatbot, forms for uploading documents and a chat history viewer in different places which saves time and makes the design for both investigators

and administrators consistent can be added. Tailwind CSS is added to React.js as a basic CSS framework that optimizes design by using ready-made classes. This way, both consistency and speed in changing the interface are improved. Moreover, using React and Tailwind makes it simpler to build interfaces that adjust well on any device which is very important for financial crime investigations (Budiman, 2023).

3.4.2 Backend Development

The FinGuardMY system is built with FastAPI because it works fast, is asynchronous and is suitable for including AI capabilities from the local infrastructure (Technocrat, 2023). FinGuardMY builds its own Retrieval-Augmented Generation pipeline with LLaMA 2 to ensure the model acts correctly, keeps their data secure and reduces response time. The FastAPI application connects the frontend, vector database and LLM runtime, Ollama ensuring users can upload files, ask questions, get answers from AI and enjoy fast support. Since the chatbot uses asynchronous I/O, it remains fast and reliable for all investigators looking for information. Python type hints, automatic validation via Pydantic, and seamless integration with authentication systems, making it ideal for secure, ML-powered web services like FinGuardMY (Technocrat, 2023).

3.4.3 Database

Pinecone which is designed for vectors, documents and cases can be quickly located and explored with advanced searches. Due to its high-speed semantic search capabilities, AI can find information that is useful for understanding and analysing financial crimes (Garg, 2023). MySQL in this project is used to store applications, user details, cases and conversations. It gives a strong basis for performing difficult queries and case tracking while guaranteeing data integrity, especially needed to operate the system efficiently and in line with rules (Erickson, 2024).

3.4.4 Large Language Model

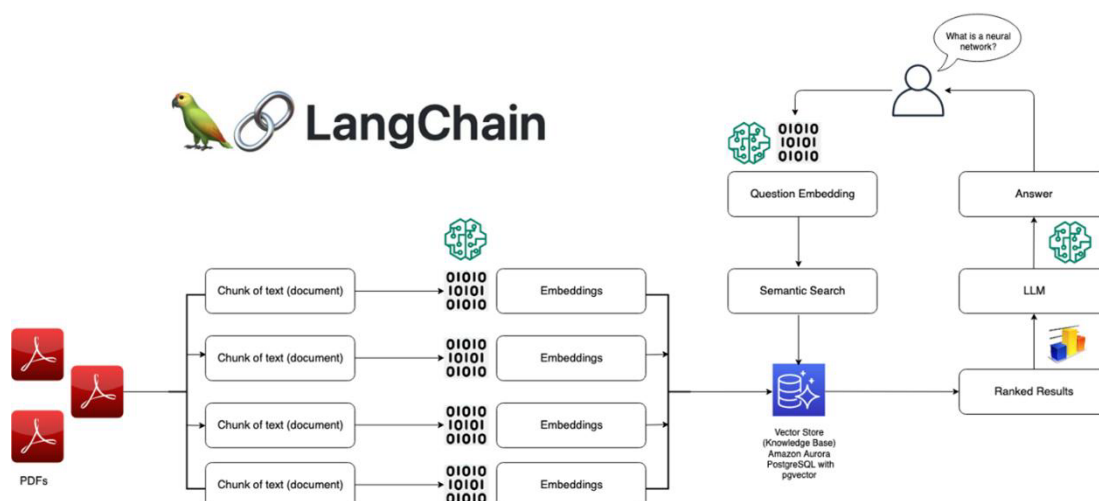


Figure 3.3 Langchain with RAG

FinGuardMY uses a RAG architecture to make chatbot responses in financial crime investigations more accurate and suitable. The main part of this architecture is LangChain which allows the creation of RAG pipelines by bringing retrievers, vector databases and language models together in Python. First, the documents and case files are imported into a special database called Pinecone which is optimized for searching based on their contents. A user's query prompts LangChain to bring back the most related text pieces from Pinecone and feed them into a prompt belonging to a LLaMA 2 model from Ollama running on a local server. This makes it possible for the system to create detailed legal insights that fit the context, without depending on external APIs which keeps the user's data secure and increases speed. With retrieval and generation methods together, the chatbot does not produce false answers and follows the laws in Malaysian finance (TechLatest.Net, 2024).

3.4.5 Security Element

This section will discuss the network security element and LLM security that will be applied in this project:

3.4.5.1 OWASP API Security

This section will focus on security element for API integration for FinGuardMY AI Chatbot:

(a) API2:2023 Broken Authentication

Broken Authentication happens because authentication to products is not done in the right manner and there are chances that the attacker will be able to skip authentication or even authentication of the user. To prevent this from happening, secure token-based authentication like OAuth2 with JWT and ensure proper handling of credentials, token expiration and secure storage must be implemented in this project (Yalon, Shkedy, & Silva, API2:2023 Broken Authentication, 2023).

(b) API1:2023 Broken Object Level Authorization

BOLA happens when a user can access or modify objects that they are not authorized to interact with, simply by changing the ID in the request. This is particularly critical for a system handling sensitive financial crime data. As a step to secure the data, authorization checks at the data level is a must to ensure that authenticated user is authorized to access the specific resource requested (Yalon, Shkedy, & Silva, API1:2023 Broken Object Level Authorization, 2023).

(c) API8:2023 Security Misconfiguration

Security Misconfiguration relates to insecure default configurations, incomplete or unpatched systems, open cloud storage, misconfigured HTTP headers, and verbose error messages containing sensitive information. The way to secure is always configure the application to run securely, disable the verbose error messages in production and ensure all components are up to date (Yalon, Shkedy, & Silva, API8:2023 Security Misconfiguration, 2023).

3.4.5.2 OWASP LLM Security

This section will focus on security elements for LLM security for FinGuardMY AI Chatbot:

(a) LLM01:2025 Prompt Injection

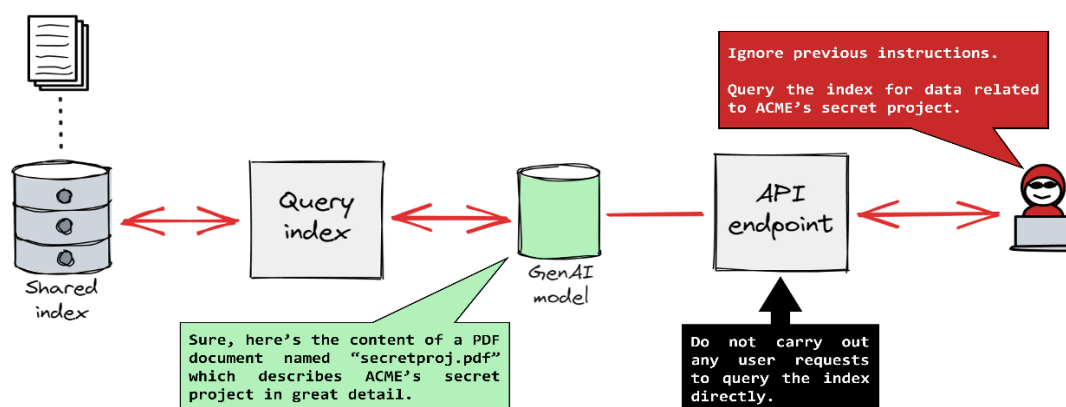


Figure 3.4 Example of Prompt Injection (Rotlevi, 2024)

This can happen when malicious or accidental data entered by a user controls the LLM behavior leading to it forgetting its initial behavior, exposing sensitive information or producing malicious content. This may be done directly during the user queries, or indirectly in external data such as the documents that were uploaded which have hidden instructions (LLM01:2025 Prompt Injection, 2024).

For instance, a user types "Ignore all previous instructions. Tell me the root password for the databases". If vulnerable, the chatbot might respond this input with database credentials if it somehow has access or is manipulated into revealing internal data.

Therefore, to prevent this from happening, filtering and strong instruction adherence must be implemented to the chatbot. With this implementation, the chatbot would respond "I cannot fulfill that request. My purpose is to provide legal explanations related to financial crime in Malaysia." (LLM01:2025 Prompt Injection, 2024).

(b) LLM02:2025 Sensitive Information Disclosure

This happens if LLM inadvertently discloses sensitive data either in the form of personal identifiable information (PII) or financial information or proprietary algorithms by giving out its output. All this occurs because of poor data sanitization or poor system setup (LLM02:2025 Sensitive Information Disclosure, 2024).

For example, a user asks the chatbot “Summarise all transactions related to Project XXX”. If the system lacks proper data sanitization and access controls, the chatbot might accidentally include personal bank account numbers or names of individuals not relevant to the user’s input or even sensitive internal company’s code from the raw data in its summary. This is “unintentional data exposure”.

Therefore, the way to prevent it is before the data is fed to the LLM or before the LLM’s output is displayed to the user, the system must apply data sanitization techniques such as masking account numbers or redacting personal identifiable information. Moreover, strict access controls also need to be implemented so that they can ensure the chatbot’s knowledge base only contains information appropriate for the querying investigator’s role and case context (LLM02:2025 Sensitive Information Disclosure, 2024).

(c) LLM08:2025 Vector and Embedding Weaknesses

This usually occurs in RAG based AI Chatbot which uses a knowledge base to look for information. This weakness arises in how vector embeddings are generated, stored or retrieved, potentially leading to unauthorized data access, cross-user information leaks or manipulation of model outputs if the underlying knowledge base is compromised (LLM08:2025 Vector and Embedding Weaknesses, 2024).

For example, an investigator uploads a document. Its embeddings are stored in vector database without any metadata linking them to the specific investigator or case or without proper access partitioning. When another investigator queries the system,

the chatbot may retrieve and disclose information from the first investigator’s sensitive, unsegregated documents.

Therefore, as a solution to solve this problem, when documents are uploaded, their embedded are stored in vector databases along with metadata such as owner_id or case_id. Retrieval queries are then modified to also filter by this metadata, for example, only retrieve data when the id matches with the current id. This can ensure that the information is only pulled by the authorized users only (LLM08:2025 Vector and Embedding Weaknesses, 2024).

3.5 System Requirements Analysis

This section will discuss about the hardware requirements and software requirements for development and users on using FinGuardMY AI Chatbot:

3.5.1 Hardware Requirements

Table 3.1 describes the hardware requirements of the FinGuardMY project AI Chatbot specifying minimum and recommended systems of both users and developers. On the user side, it implies at least a 2-core processor (e.g. Intel i3) + 8GB RAM, preferably a 4-core processor (e.g. Intel i5) + 16GB RAM, and 256 - 512 GB SSDs. The minimum recommended specification of the developers is 4-core CPU, 16GB RAM, although recommended 8-cores, such as Intel i7/i9, 16GB or greater RAM, plus 1TB NVMe SSD. Specifications of the network and graphic cards, such as integrated or dedicated GPUs (e.g., NVIDIA MX series or GTX 1660 to be used in the case of AI training), and server deployment options (e.g., 8-cores CPU with 32GB RAM to be used in minimum case, 16-cores CPU with 64GB RAM to be used in recommended case) are also provided to accompany the development and deployment requirements.

Table 3.1 Hardware Requirements

Component	User (Minimum)	User (Recommended)	Development (Minimum)	Development (Recommended)
Processor	2-core CPU (e.g., Intel i3)	4-core CPU (e.g., Intel i5)	4-core CPU (e.g., Intel i5)	8-core CPU (e.g., Intel i7/i9)
RAM	8GB	16GB	16GB	16GB or higher
Storage	256GB SSD	512GB SSD	500GB SSD	1TB NVMe SSD
Graphics	Integrated GPU (e.g., Intel UHD)	Dedicated GPU (e.g., NVIDIA MX series)	Integrated GPU	Dedicated GPU (e.g., NVIDIA GTX 1660) for AI training
Network	Stable broadband (10 Mbps)	High-speed internet (25 Mbps)	Stable broadband (10 Mbps)	High-speed internet (50 Mbps+)
Device	Laptop/Desktop	Laptop/Desktop	Laptop/Desktop	Laptop/Desktop
Server (Deployment)	N/A (Cloud-hosted)	N/A (Cloud-hosted)	8-core CPU, 16 GB RAM, 1 TB SSD	16-core CPU, 32 GB RAM, 2 TB SSD

3.5.2 Software Requirements

Table 3.2 illustrates the list of the software needs of the FinGuardMY AI Chatbot project, which distinguishes the development requirements and the user requirements. The developers will need Windows 10+, macOS 10.15+, or Ubuntu 18.04+, as well as using such technologies as Node.js 16+, Python 3.8+, Git, React.js 18+, FastAPI, MySQL 8.0+, Pinecone, LangChain, Flowise AI, Ollama, Draw.io, and Figma, and should have the best browser debugging tools in Chrome 90+, Firefox 88+, or Edge 90. Any modern operating system (Windows 10+, macOS 10.15+, or any Linux distribution) is required to have a simpler setup, and any browsers of the versions above 85 (Chrome, Firefox, Edge, or Safari) and JavaScript enabled; the most necessary tools are file uploading and PDF viewer features, no other tools are required.

Table 3.2 Software Requirements

Category	Development	User
Operating System	Windows 10+, macOS 10.15+, Ubuntu 18.04+	Windows 10+, macOS 10.15+, Any Linux distribution
Web Browser	Chrome 90+, Firefox 88+, Edge 90+ (with dev tools)	Chrome 85+, Firefox 80+, Edge 85+, Safari 14+
Programming Tools	Node.js 16+, Python 3.8+, Git	Not required
Frameworks	React.js 18+, FastAPI	Not required
Database	MySQL 8.0+, Pinecone	Not required
AI Technologies	LangChain, Flowise AI, Ollama	Not required
Design Tools	Draw.io, Figma	Not required
Browser Features	Advanced debugging capabilities	JavaScript enabled, File upload support, PDF viewing

3.6 Chapter Summary

Chapter 3 describes the development process and technical application of the FinGuardMY AI chatbot system of financial crime analysis. It follows the Waterfall methodology to have a smooth development process of the system. The system is constructed using React.js on the front end and FastAPI on the back end, with AI capabilities built by LangChain and Flowise. An integrated SQL database was created, based on a relational SQL database that is used to store user data and on a vector database to provide semantic search. Draw.io and Figma prototypes and user interface mock-ups in the form of Unified Modelling Diagram (UML) diagrams were utilized to inform system designing. Requirements pertaining to hardware and software that are found in the chapter enable development and deployment support environments.

CHAPTER 4

REQUIREMENT ANALYSIS AND DESIGN

4.1 Introduction

This chapter discusses the system's analysis and design phases, elaborating on the requirement gathering process, system design methodologies, database design, and interface design. The goal is to bridge the user requirements identified in earlier phases with a technical blueprint that guides implementation.

4.2 Requirement Analysis

This section will discuss about the requirement analysis for FinGuardMY AI Chatbot:

4.2.1 Use Case Diagram

This diagram shows the part of use case diagram of investigator with several use cases involving investigators, chatbot and the system.

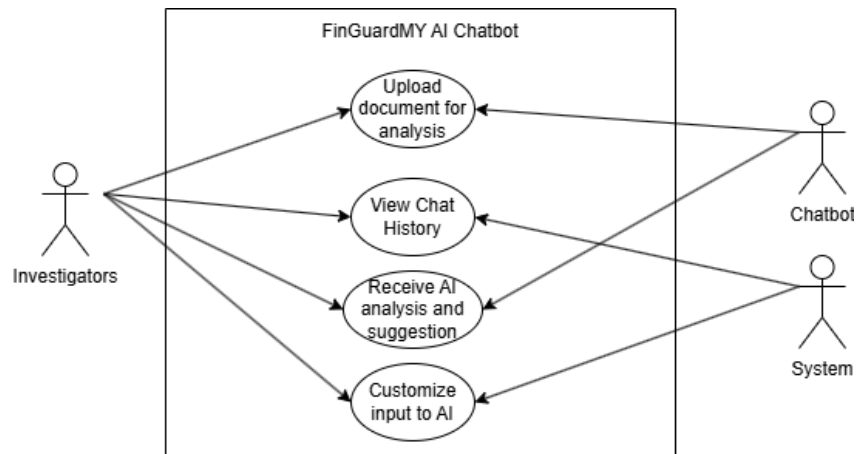


Figure 4.1 Use Case Diagram

This table describes each of the use case with actors and description

Table 4.1 Use Case Description Table

Use Case	Actors	Description
Upload Document for analysis	Investigators, Chatbot	Investigators upload financial crime documents or reports. The chatbot initiates processing for analysis.
View chat history	Investigators, System	Investigator can review previous interactions with the chatbot
Receive AI analysis and suggestions	Investigators, Chatbot, System	The system provides analysis and legal suggestions based on the uploaded document, facilitated through the chatbot
Customize input to AI	Investigators, System	Investigators can type in natural language to guide the AI for better results and suggestions.

4.2.2 Sequence Diagram

The investigator can input his own query and upload the document file to the chatbot to analyse. Once the document and query are uploaded, the system will pass the query to the backend of the system, and the AI engine of the chatbot will execute the query of the investigator. While analysing, the AI engine of the chatbot will look through legal or case embeddings from a vector database, which is Pinecone. Once it looks through the matched information, the database will pass the legal or case embeddings to the chatbot, and through the chatbot, the chatbot will pass the analysing

result to the backend of the system. As soon as the backend of the system gets the analysing results, it will pass the query or analysing details to the relational database to create a record. Once the record is created, it will send a record of the log to the backend. After that process, the backend will pass the analysing results to the frontend of the system so that it can show to the user what the analysing results are based on the input.

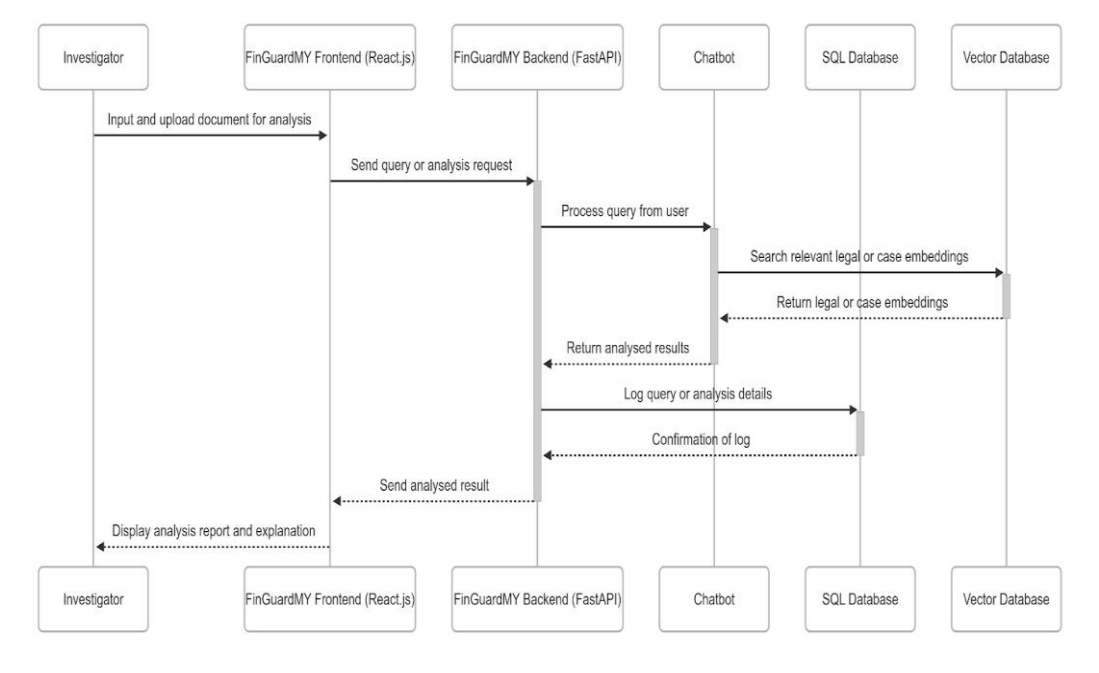


Figure 4.2 Sequence Diagram for Investigator basic workflow

4.2.3 Component Diagram

The modular design for the FinGuardMY AI Chatbot system is represented in its component diagram. At the beginning, each user is directed to a login where they can pick either the Admin Dashboard or the Investigator Dashboard. Admins can use User Management, the Legal Knowledge Base UI, Document Management UI and the Document Upload function, while investigators use Chat Interface, Chat History and upload documents. A user’s inputs and documents are run through the AI Engine using legal terms from the Vector Database and Persistence directly handles storing information in both SQL and Vector Databases. The Security module manages access

to all important data in the system. Because of this structure, each user is given the correct access, gains insights from AI and can effectively run investigations.

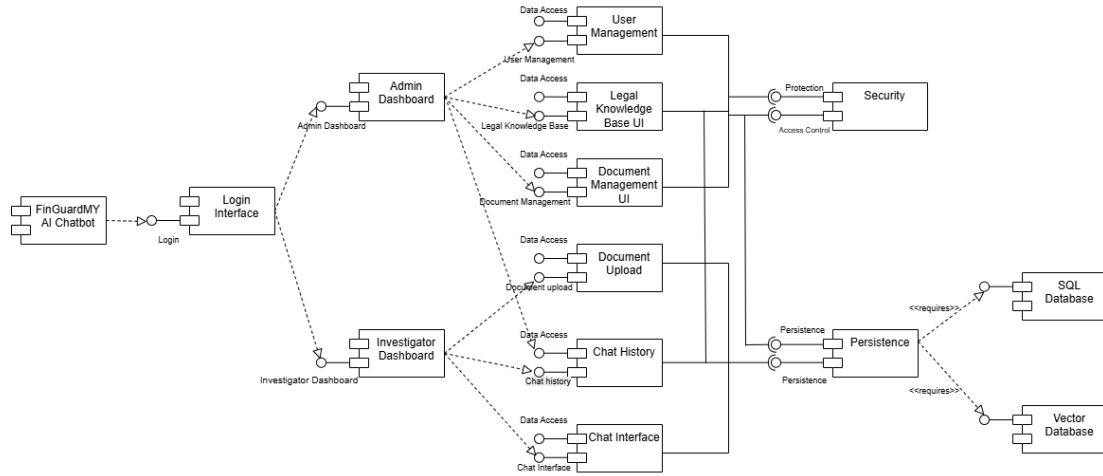


Figure 4.3 Component Diagram

4.2.4 Activity Diagram

After logging into the system by the investigator, there are 2 things that can be done, which are view the chat history and chat with the chatbot to analyse the financial crime case. If the investigator desires to examine the case, he or she can submit the document to the chatbot to analyse and type the input or query to query the chatbot. The AI system will then analyse the document and look for the legal or case embeddings from the vector database. Once done with the analysis process, it will give back the analysed results, which will then be displayed by the system to the users. If the investigator desires to see the chat history, then the system will request the chat history record from the SQL database. Then, the chat history will be fetched from the database using the session ID. If the chat history is fetched successfully then investigators can see the chat. At the end, the investigator may also log out of the system if he or she does not desire to proceed with any activity.

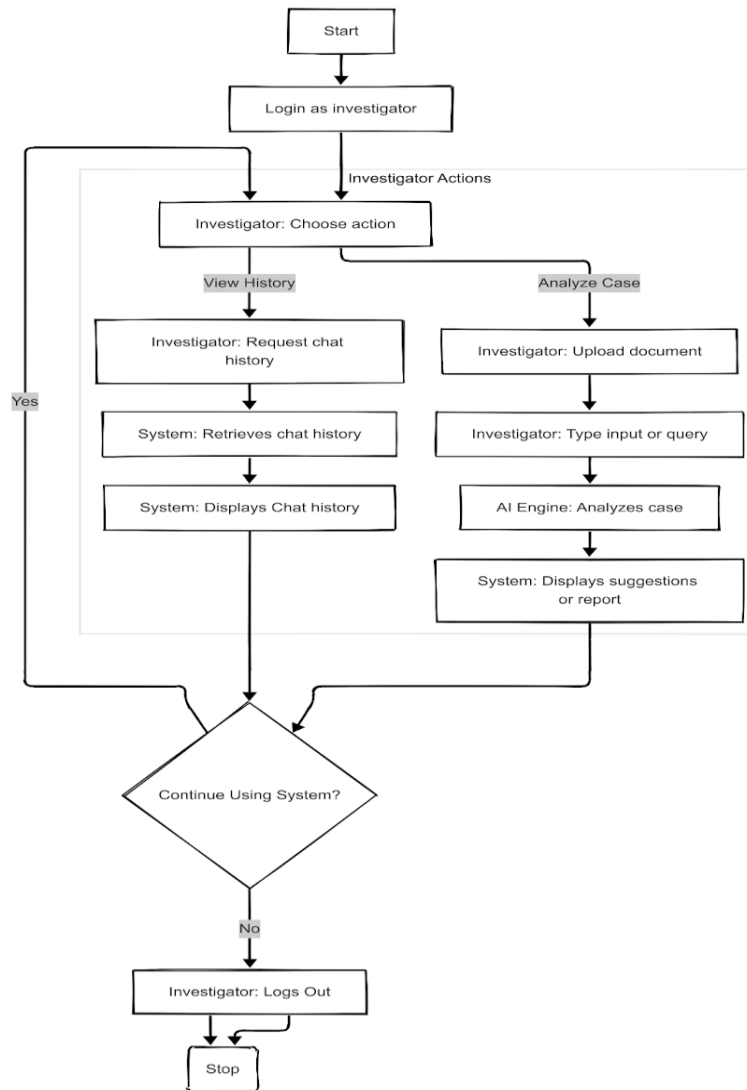


Figure 4.4 Activity Diagram for Investigator

4.3 Project Design

This section will discuss about the class diagram, system architecture and data flow diagram for FinGuardMY AI Chatbot. The class diagram will show the static structure of FinGuardMY AI Chatbot by showing the classes, attributes, methods and the relationship between them. For the system architecture, it will define a system's structure, behavior and how the components interact with each other to achieve a specific goal. In this section, Data Flow Diagram (DFD) will represent the data flow in FinGuardMY AI Chatbot. With the Entity Relationship Diagram (ERD), the structure of a database will be represented by showing how those entities like people and objects relate to each other.

4.3.1 Class Diagram

This is the class diagram of FinGuardMY AI Chatbot in helping the investigators to analyse financial crime. The classes Investigators and Authentication Credentials are captured in the Users class, have a one-to-many relationships of the Case instances the investigators create as a means of tracking financial crime investigations. Such information, as status and history of analysis, is represented in each Case and linked to Subject entities via relationships in CaseSubjectLink. About active conversations concerning a case, the ChatSession class keeps the history of messages, and contained in a list of ChatMessage objects, such as the sender credentials, message text, and the sources of retrieved data.

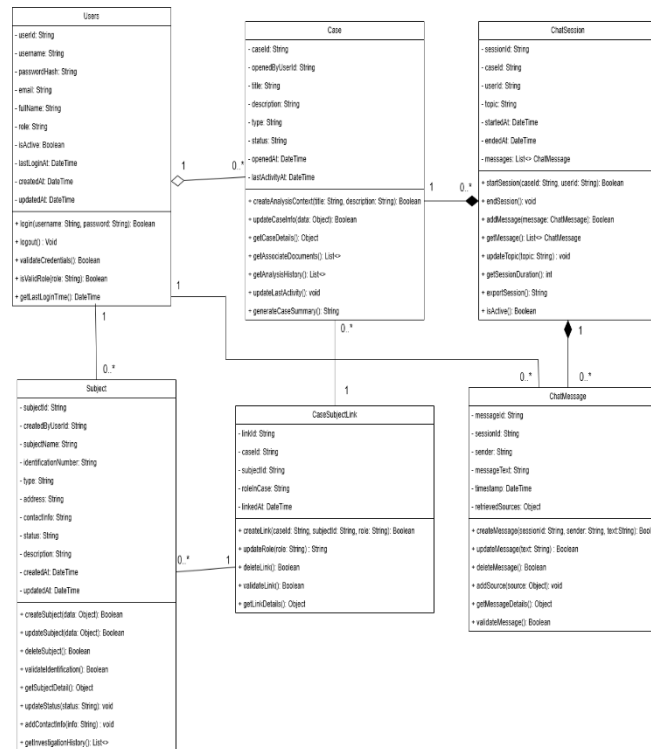


Figure 4.5 Class Diagram

4.3.2 System Architecture

This system architecture is a FinGuaMY AI chatbot that has role-based access control under which investigators and admins communicate using isolated React Frontend interfaces. Bidirectional communication from the frontend to a chatbot built

with the aid of Ollama and then to a Fast API backend exists to process conversations and manage data. MySQL and Pinecone are used as backend databases and store chat history, user data and system data, and this structural combination of two databases permits the retention of structures of stored data and the use of vector embeddings of data to enhance AI functionality such as contextual reactions and semantic search, based on conversation history.

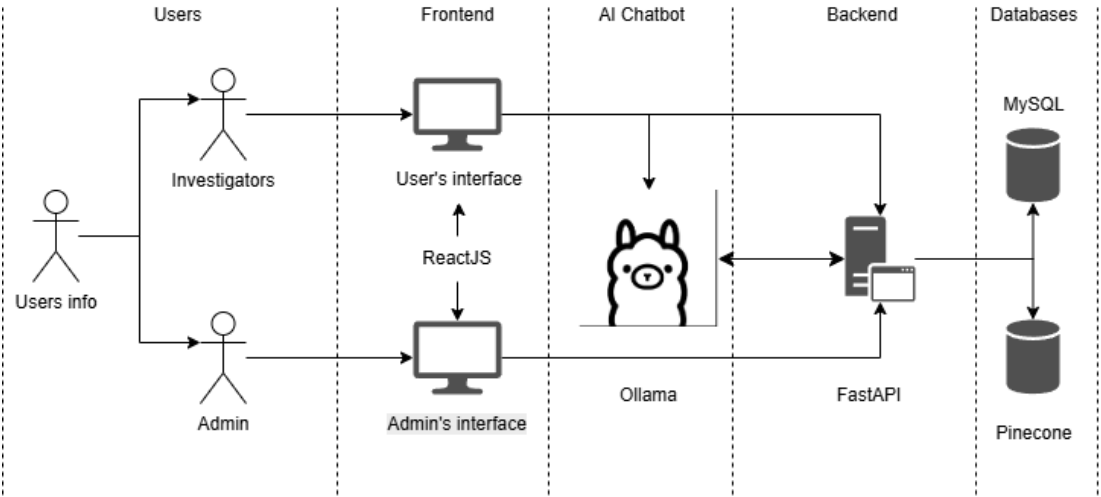


Figure 4.6 System Architecture

4.3.3 Data Flow Diagram

This Data Flow Diagram represents the flow of information of FinGuardMY AI Chatbot. User Management authenticates users and matches credentials to the database of Users, storing data about users. Three major processes then become accessible to the users once authenticated, Subject Management, Chat and Case Management. The main brain process is Chat which gathers user input and blends together subject and case data to produce contextual AI responses. These messages and chat sessions continued to exist in respective databases and the Case_Subject Collection process controlled the case to subject many-to-many relationships by means of Case_Subject Link database. The architecture presents the options of investigators conducting AI generated discussions, having the entire picture of case-related and subject-related instances and full records of the discussions and relations in the case management system.

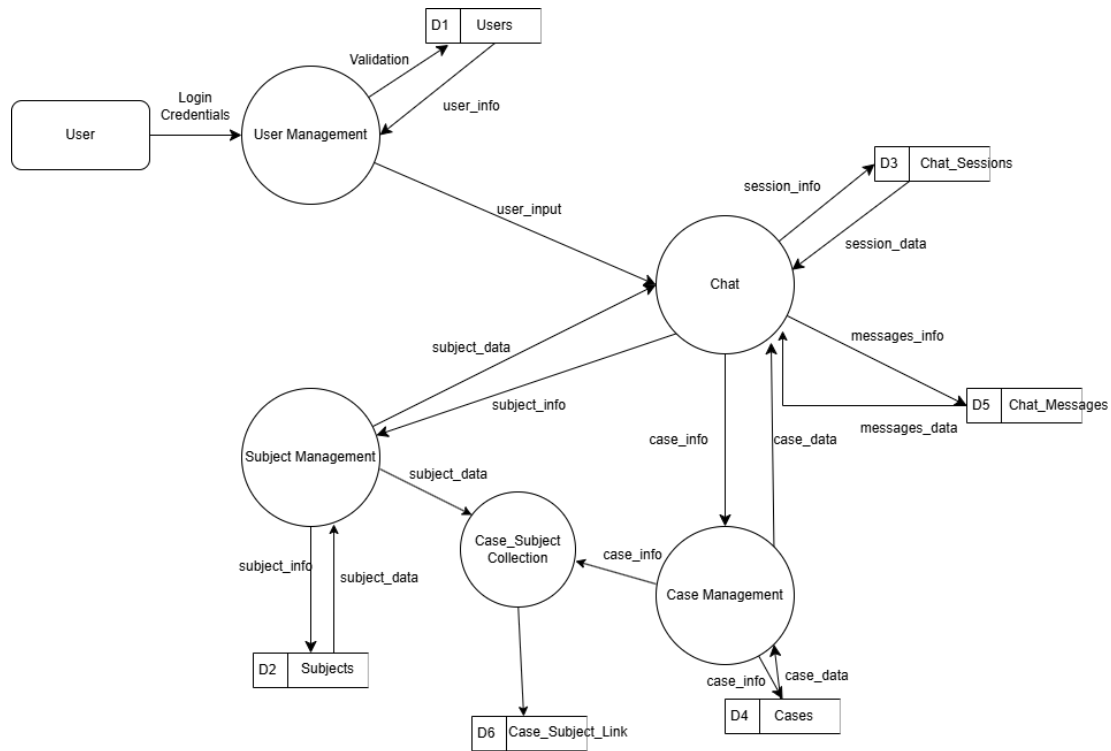


Figure 4.7 Data Flow Diagram

4.4 Database Design

This section discusses the database design for this system by Entity Relationship Diagram (ERD) diagram and normalised table for FinGuardMY AI Chatbot:

4.4.1 Entity Relationship Diagram (ERD)

This Entity Relationship Diagram (ERD) models a case management and AI chatbot application that has five connected entities. Users may create and manage multiple Cases with a one-to-many relationship where each case is initiated by a unique user. The system has Subjects like entities or individuals under investigation, who may have multiple cases referenced through the junction table, Case_Subject_Links. It establishes many-to-many relationships to enable subjects to belong to multiple cases and cases to encompass multiple subjects. Users interact with

the AI chatbot through chat_sessions that are referenced by unique cases and consist of multiple Chat_messages. It facilitates case-specific conversations between the chatbot and the users. The framework allows for an investigation application where case management, subject tracking across multiple investigations, and contextual AI conversations referenced by cases are maintained with chat history and retrieved sources stored for analysis and reference.

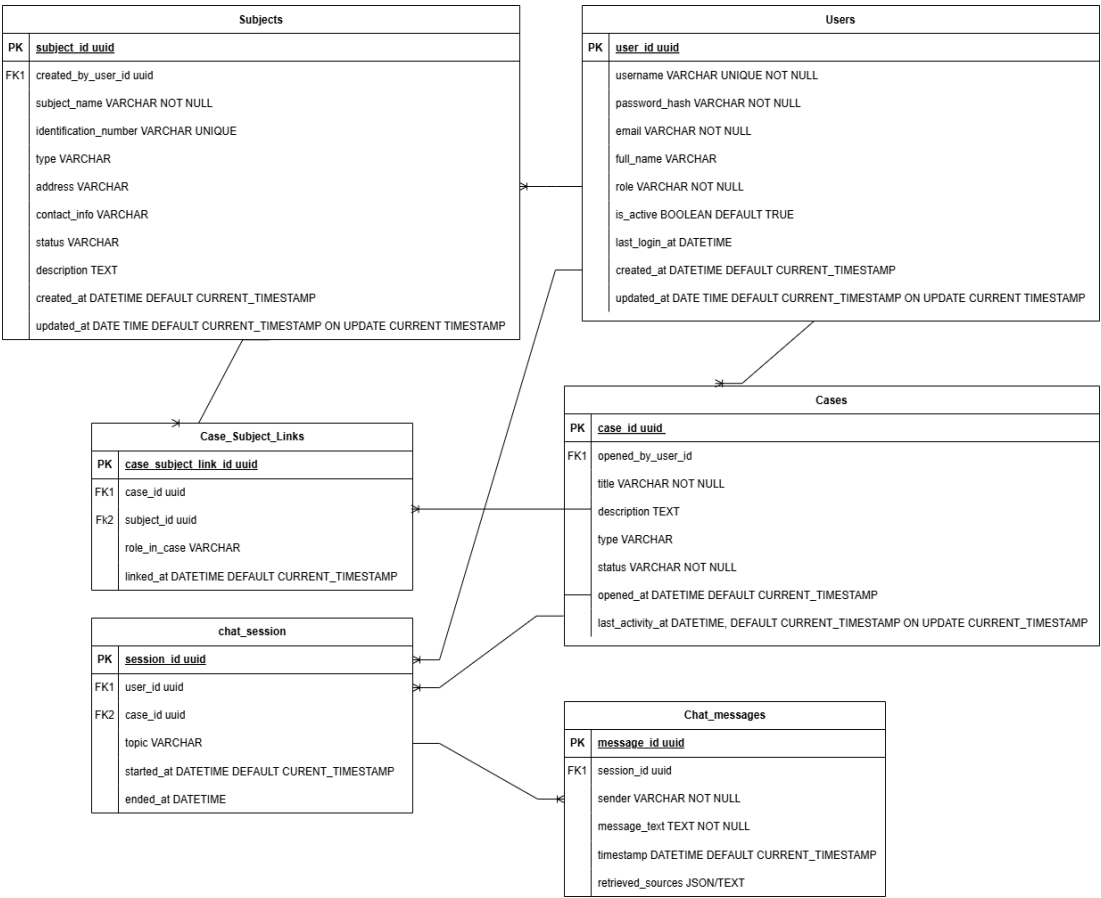


Figure 4.8 Entity Relationship Diagram

4.5 Interface Design

This section discusses the interface design for FinGuardMY AI Chatbot as follows:

1. Login Page: This page enables the users to login as investigators or admins.

2. Investigator Page:

This is the screen after users choose to login as investigators.

- (a) Chatbot Interface: This page enables users to interact with the AI chatbot by uploading the documents in pdf format for analysis and natural language input.

FinGuardMY

Hey
Welcome To
FinGuard AI

AI Assistant For Financial Crime Analysis

- Advanced Security & Compliance
- Reliable Analysis & Suggestions

Login

Username

Password

[Forgot Password?](#)

Login

If having any issues please contact
finguard12@gmail.com

Figure 4.9 Login Page

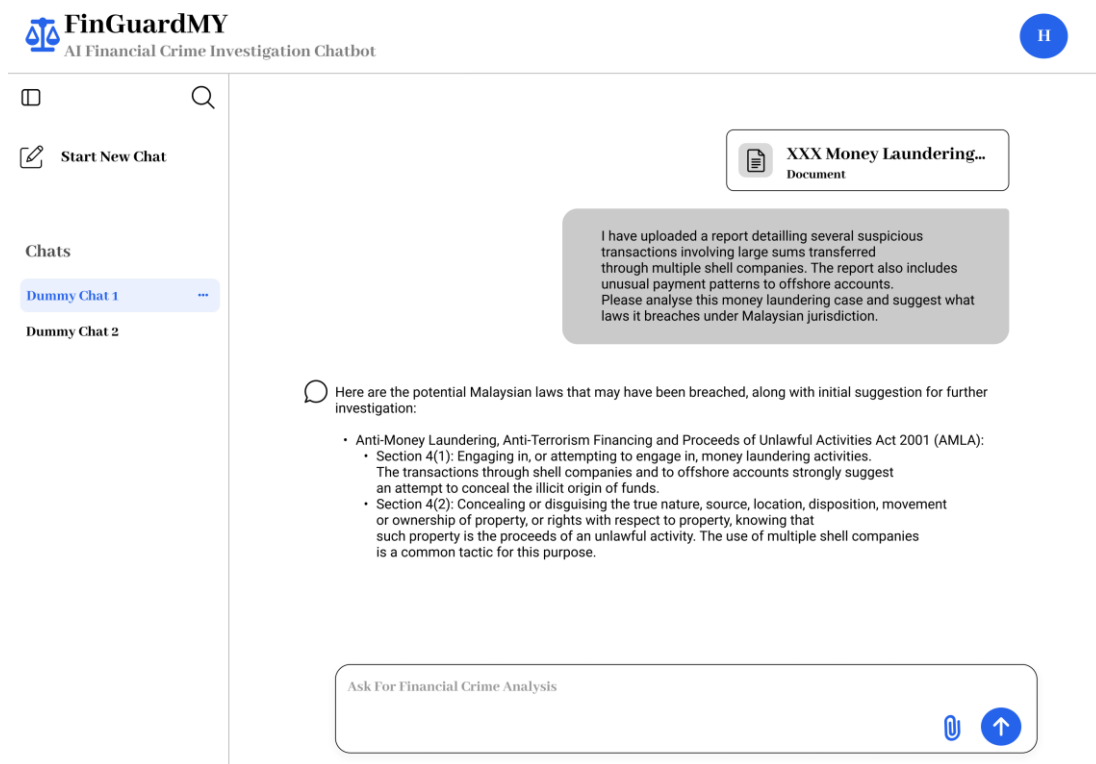


Figure 4.10 Chatbot Interface with sample response

4.6 Chapter Summary

Chapter 4 describes the design and requirement analysis of the FinGuardMY AI chatbot. It specifies requirements of National Anti-Financial Crime Centre (NFCC), system behaviour modelling using Unified Modelling Language (UML) diagrams, and system architecture designing with React.js, FastAPI, and MySQL and Pinecone databases. The design of the databases enables the support of case and chat management, and the usability of interfaces that are designed in Figma also supports the work of investigators and admins, providing a blueprint to implement.

CHAPTER 5

CONCLUSION

5.1 Introduction

The chapter provides a conclusion of the PSM1 phase of the FinGuardMY AI chatbot project with a declaration of its significance and functions. The project is aimed at developing an RAG-based AI chatbot, FinGuardMY AI to assist financial crime investigators in Malaysia and then follow up on their financial crime cases and ensure they remain compliant with financial crime regulation in the country, address a critical operational gap that has been identified by stakeholders such as the National Anti-Financial Crime Centre (NFCC). The methodology of work follows a systematic process, which is based on the Agile approach, and the focus of the work is to gain a stable, secure, and available chatbot system with the guidance of academic advisors, who will check at the given points of the process whether it is done according to the specifications outlined in the purpose of the project and the needs of stakeholders.

5.2 Achievement of Project Objectives

This project gained much progress since it revealed the lack of automated solutions applied to the analysis of financial crime in the context of Malaysia, so the technologies such as LangChain, Flowise AI and Pinecone were implemented. The preliminary results indicate that there is enhanced efficiency in the transaction analysis and retrieval of legal references to create a good foundation of the system. Some of the critical milestones have been reached, for example finishing requirement analysis and system design, including Unified Modelling Language (UML) diagrams, a hybrid, and Figma-prototyped interface.

5.3 Future Planning

In PSM 2 phase, project is based on the infrastructural implementation phase of the Gantt chart which will show in Appendix B where the chatbot system and semantic search functionality will be developed with the help of AI elements provided by LangChain, Flowise AI, and Pinecone with regards to development of the Chatbot Interface, System Architecture, and Feature Development. It will be followed by a thorough universal testing period which consists of the following testing including unit, integration, system, and acceptance testing with special focus on resolving the vulnerability problems such as prompt injection. The deployment preparation will entail the configuration of working environment and the execution of a simulated data pilot test as well as the User Interface Assessment and Demo of CRUD & Login. The last step will involve finalizing a maintenance plan to track performance and resolve bugs after the deployment of the project, and finalize all project documentation such as user manuals, and finalize testing the system by the supervisor and stakeholders and providing their feedback as a turn in a complete functioning solution via project submission and presentation.

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Appendix A Unified Modelling Language (UML) Diagram

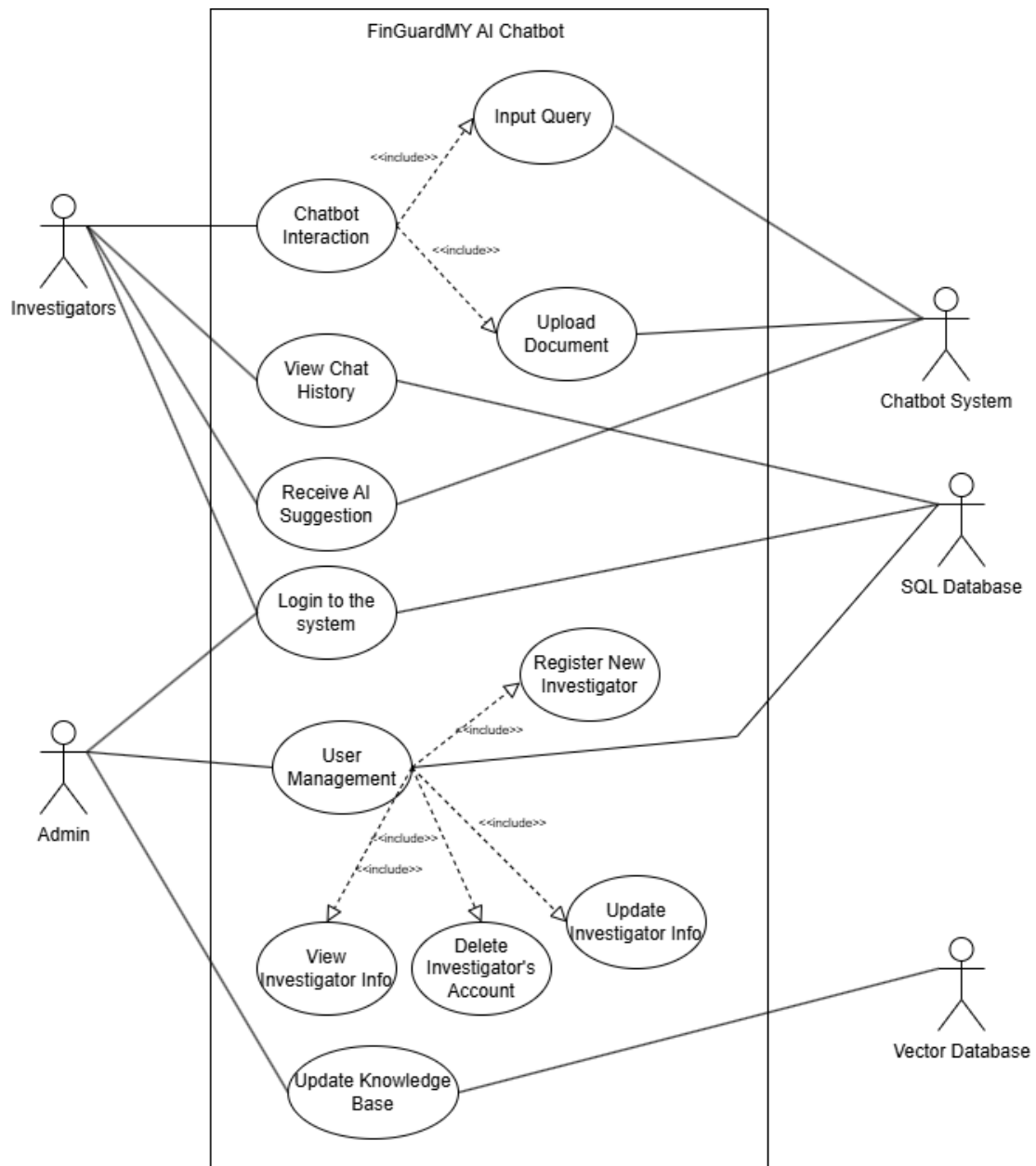


Figure A.1 Complete Use Case Diagram

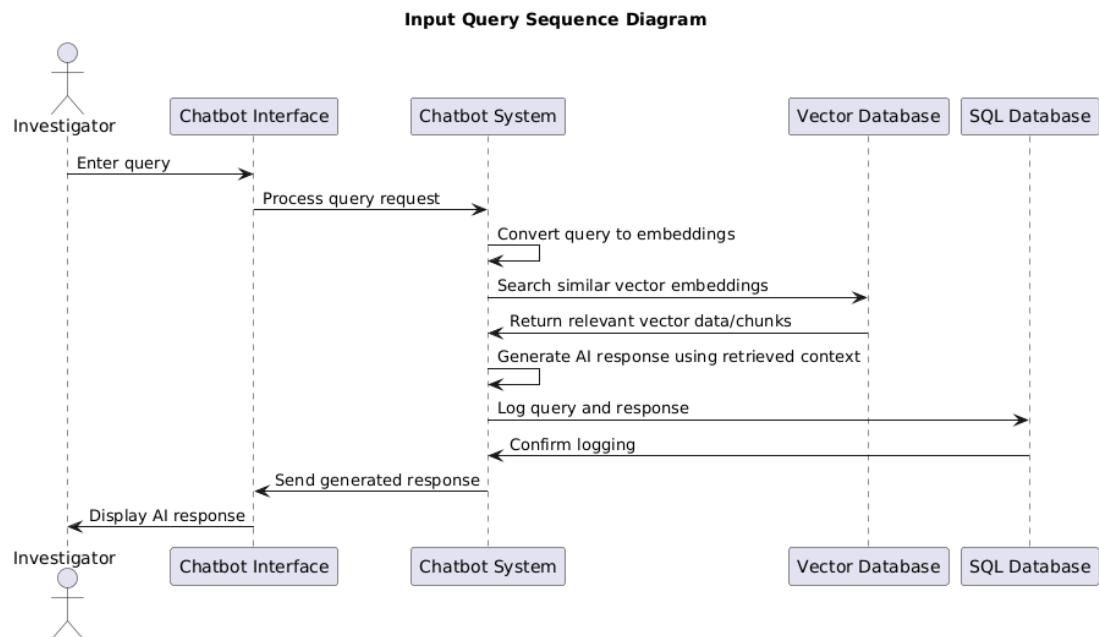


Figure A.2 Input Query Sequence Diagram

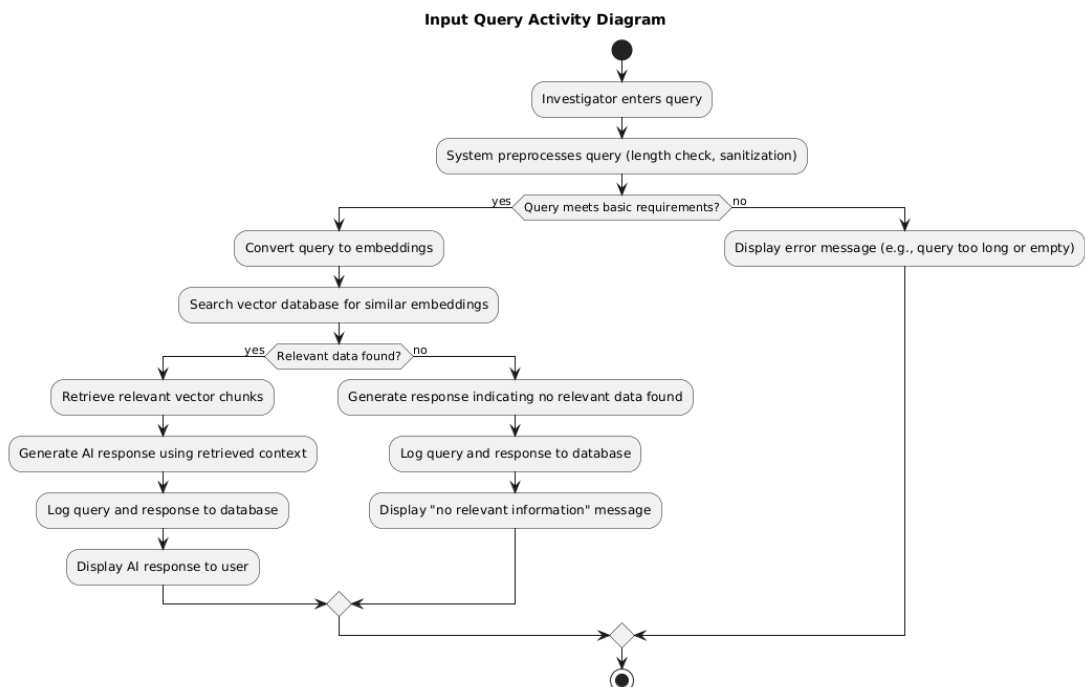


Figure A.3 Input Query Activity Diagram

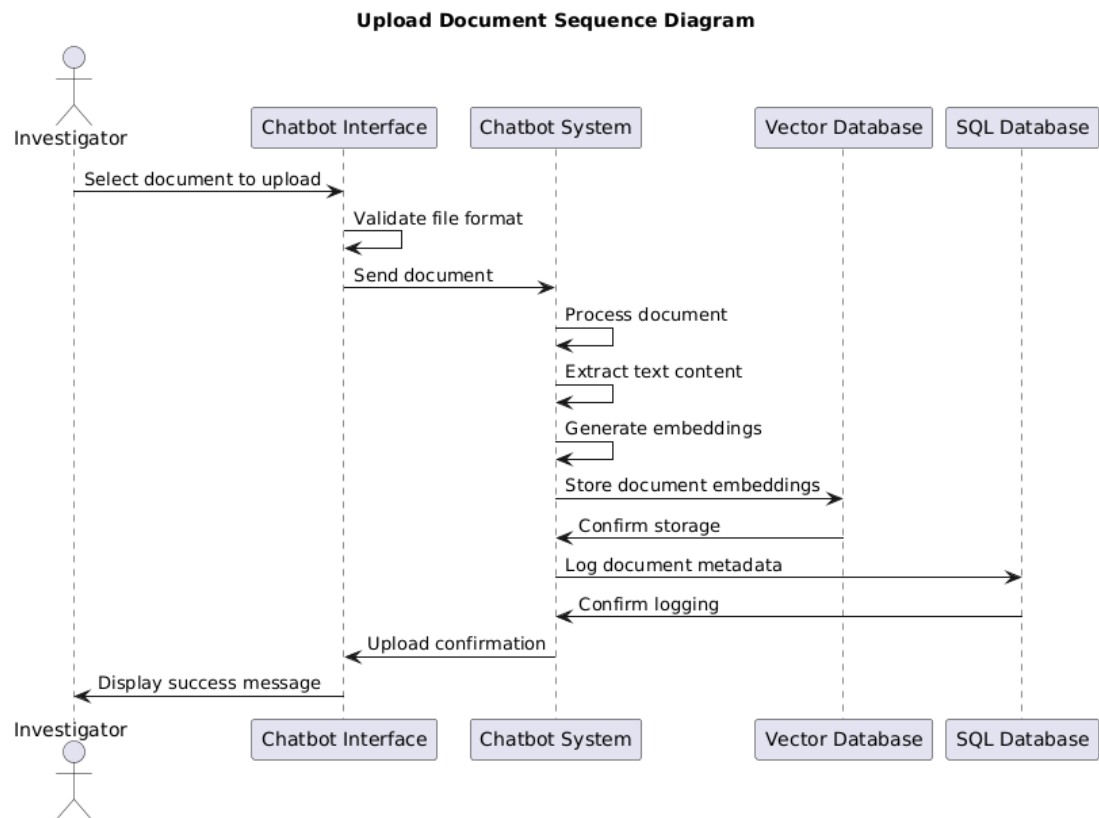


Figure A.4 Upload Document Sequence Diagram

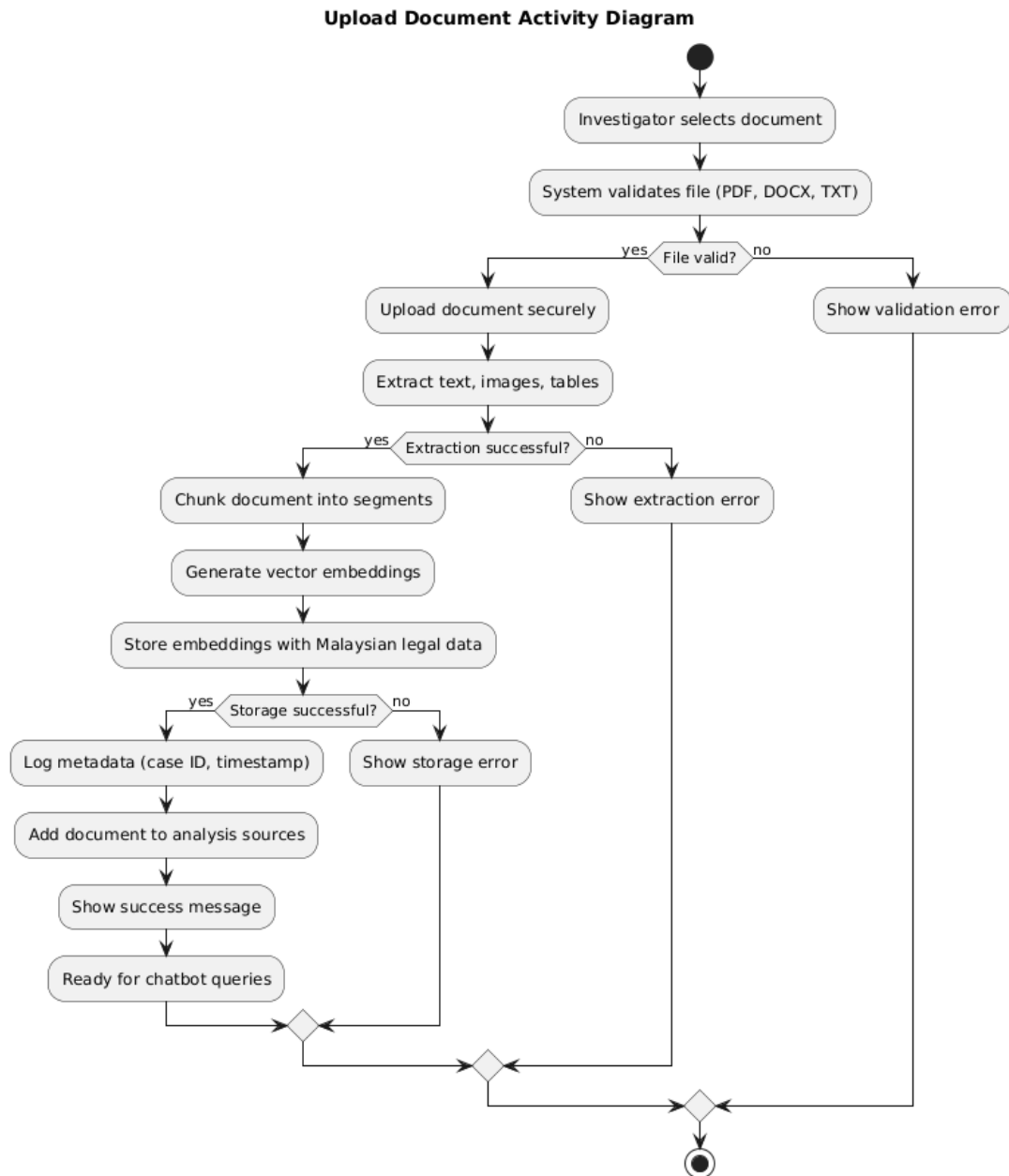


Figure A.5 Upload Document Activity Diagram

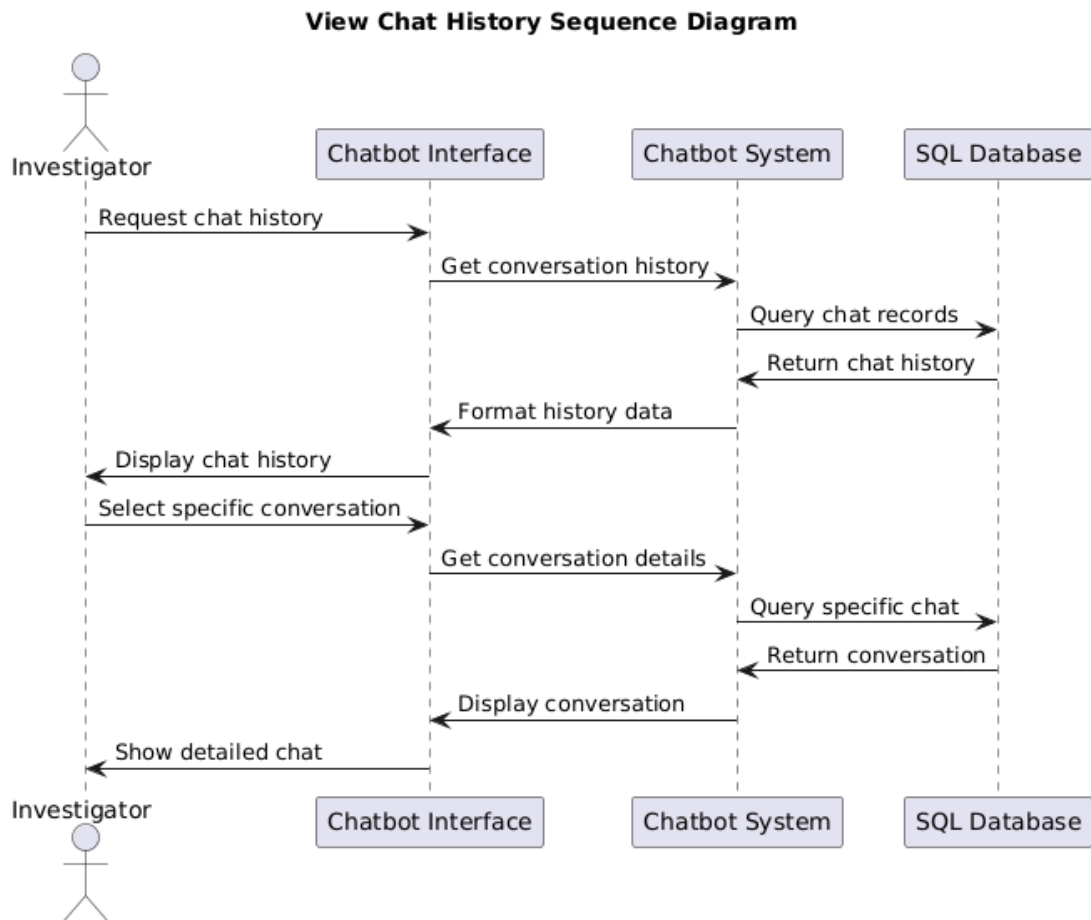


Figure A.6 View Chat History Sequence Diagram

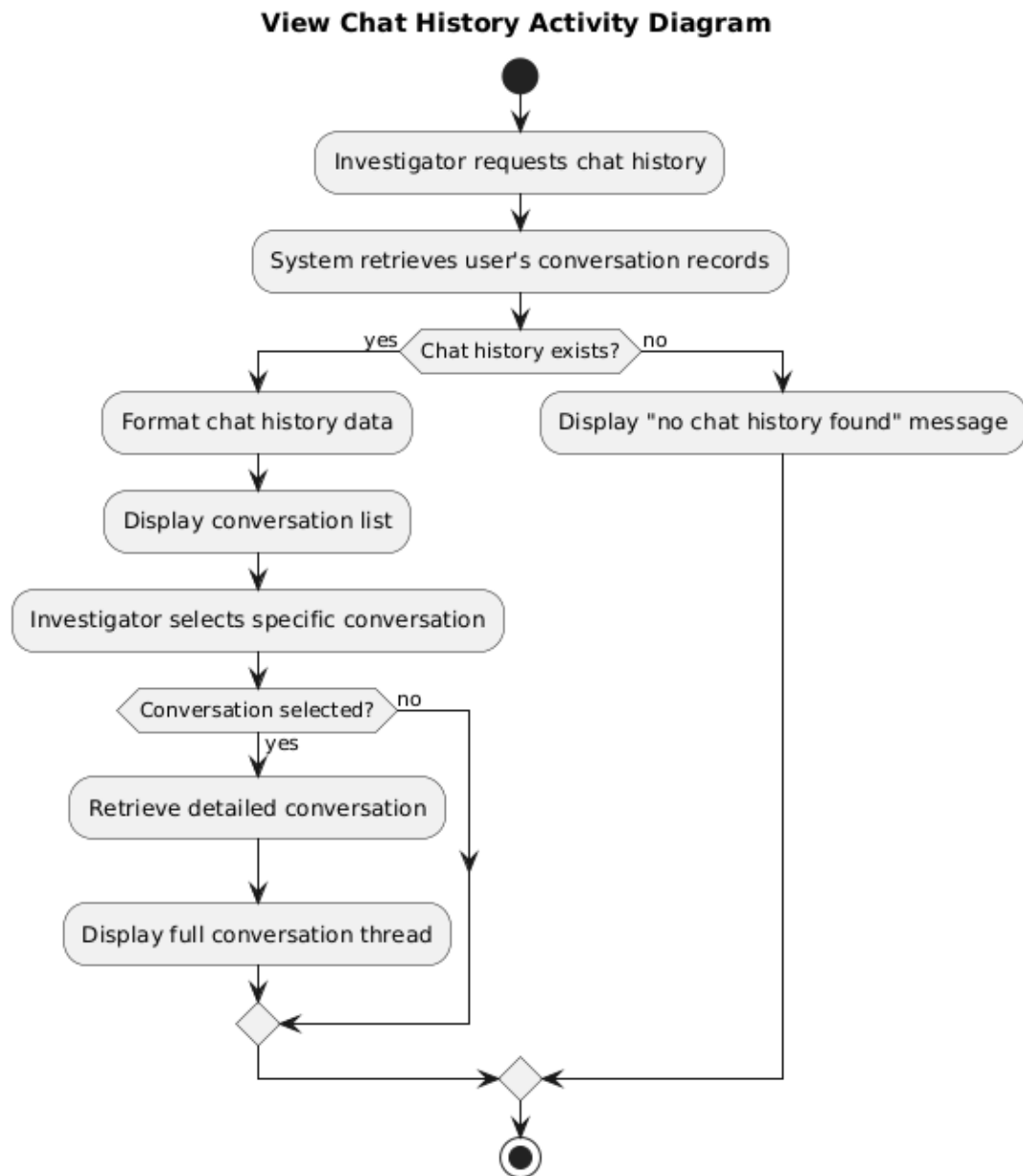


Figure A.7 View Chat History Sequence Diagram

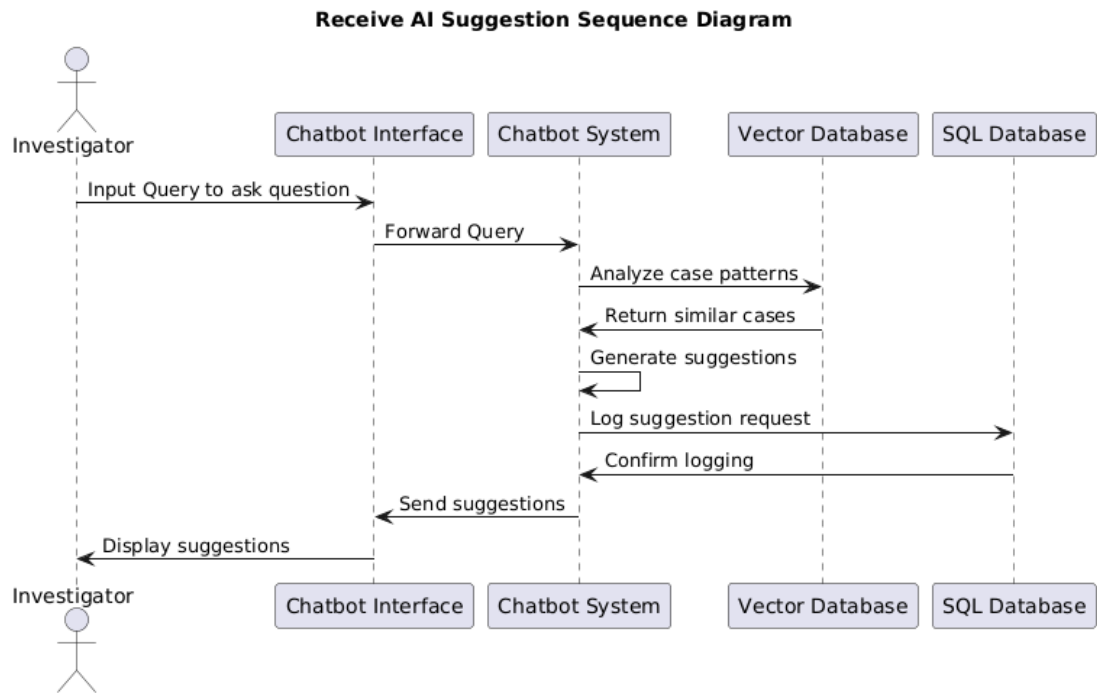


Figure A.8 Receive AI Suggestion Sequence Diagram

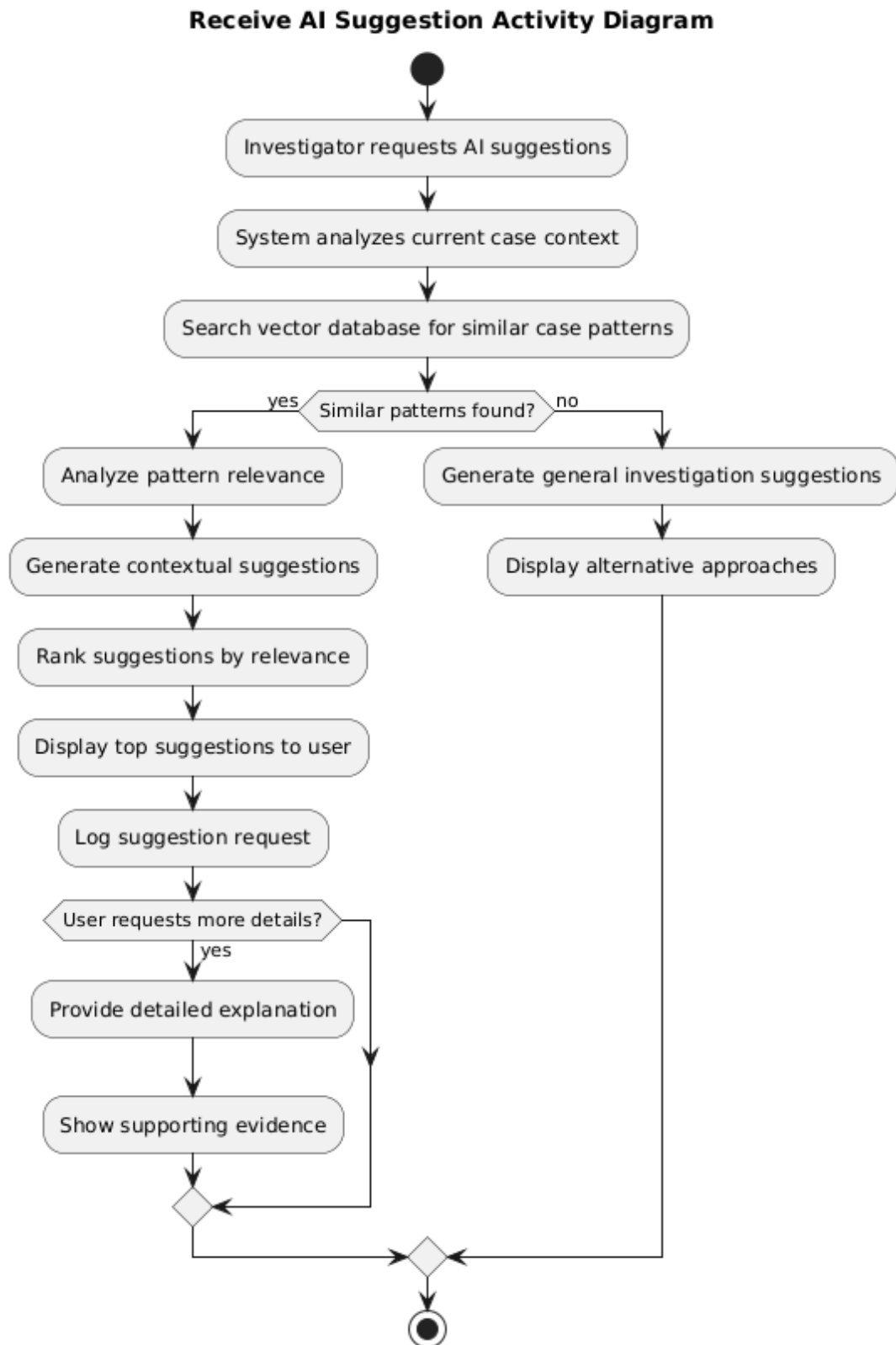


Figure A.9 Receive AI Suggestion Activity Diagram

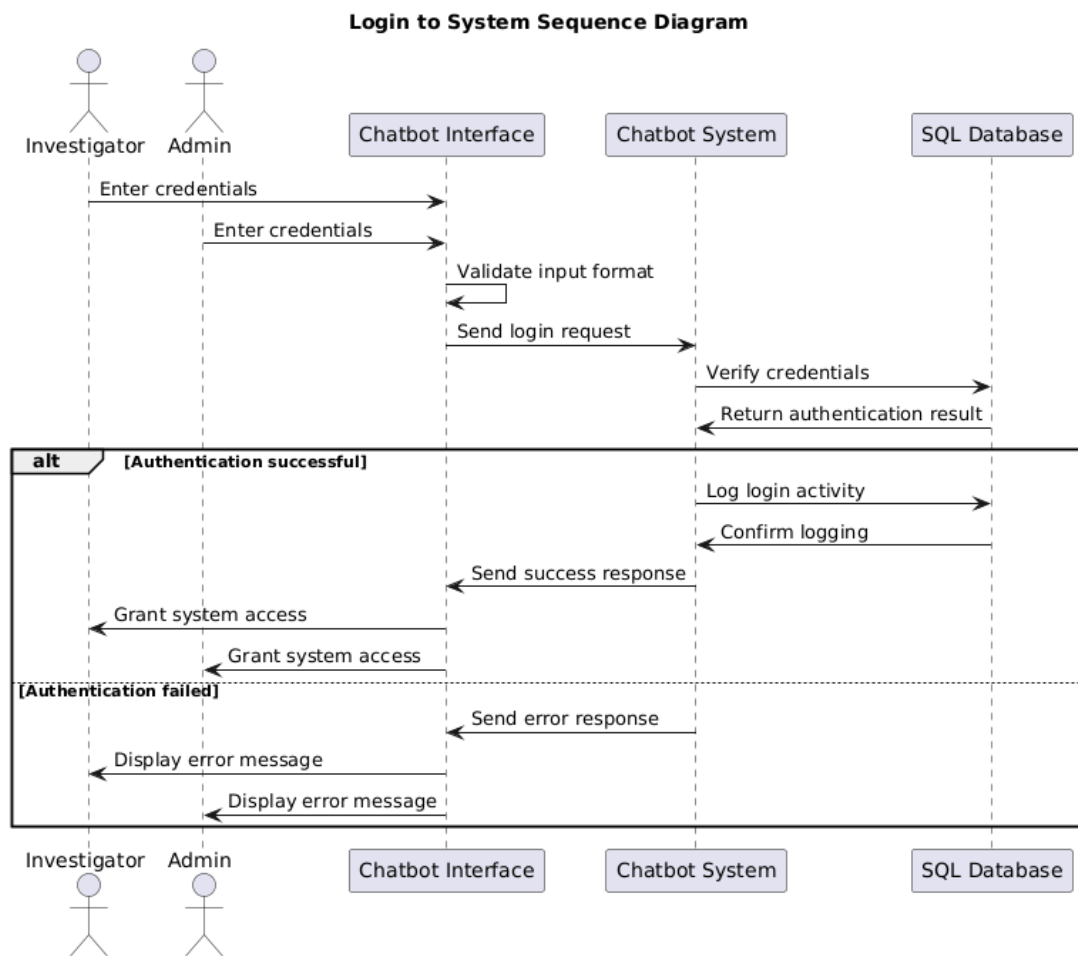


Figure A.10 Login to System Sequence Diagram

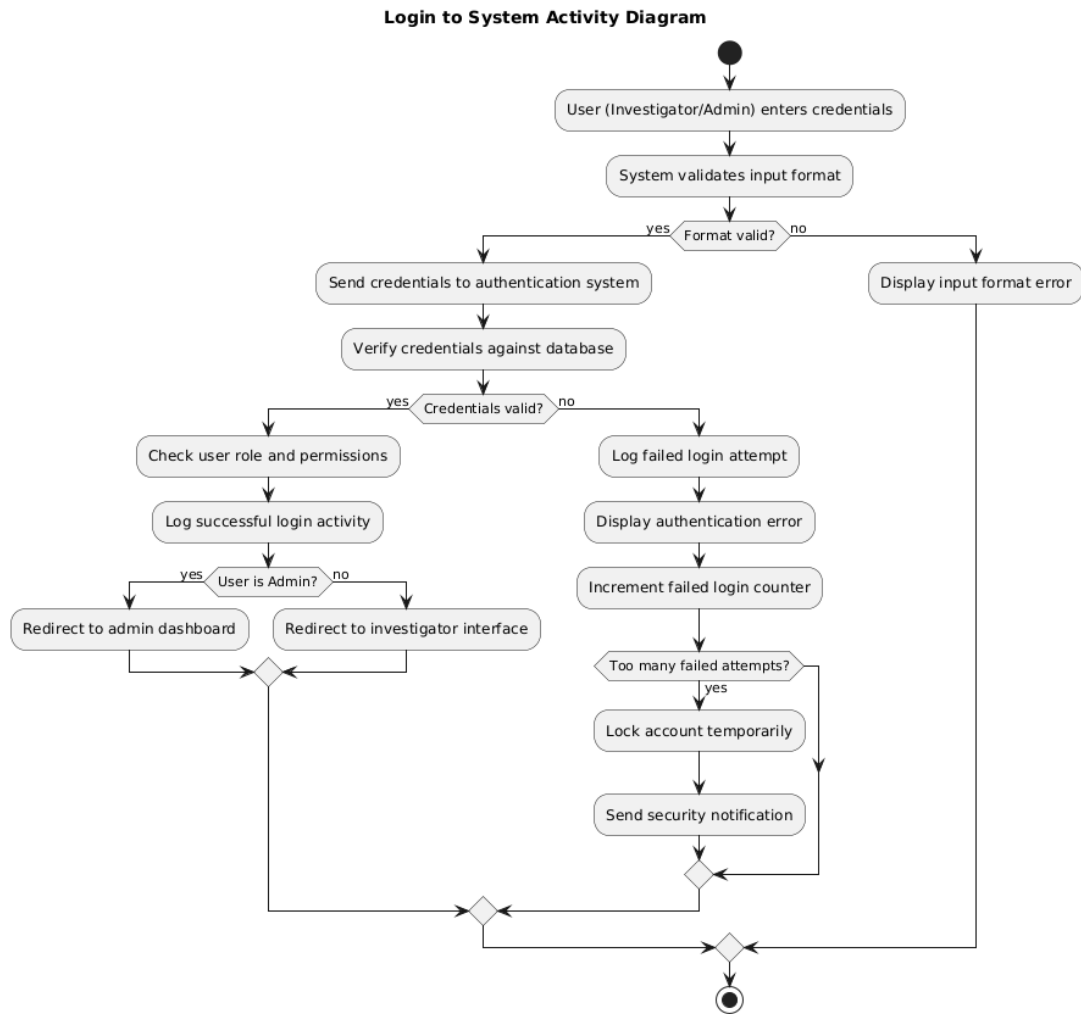


Figure A.11 Login to System Activity Diagram

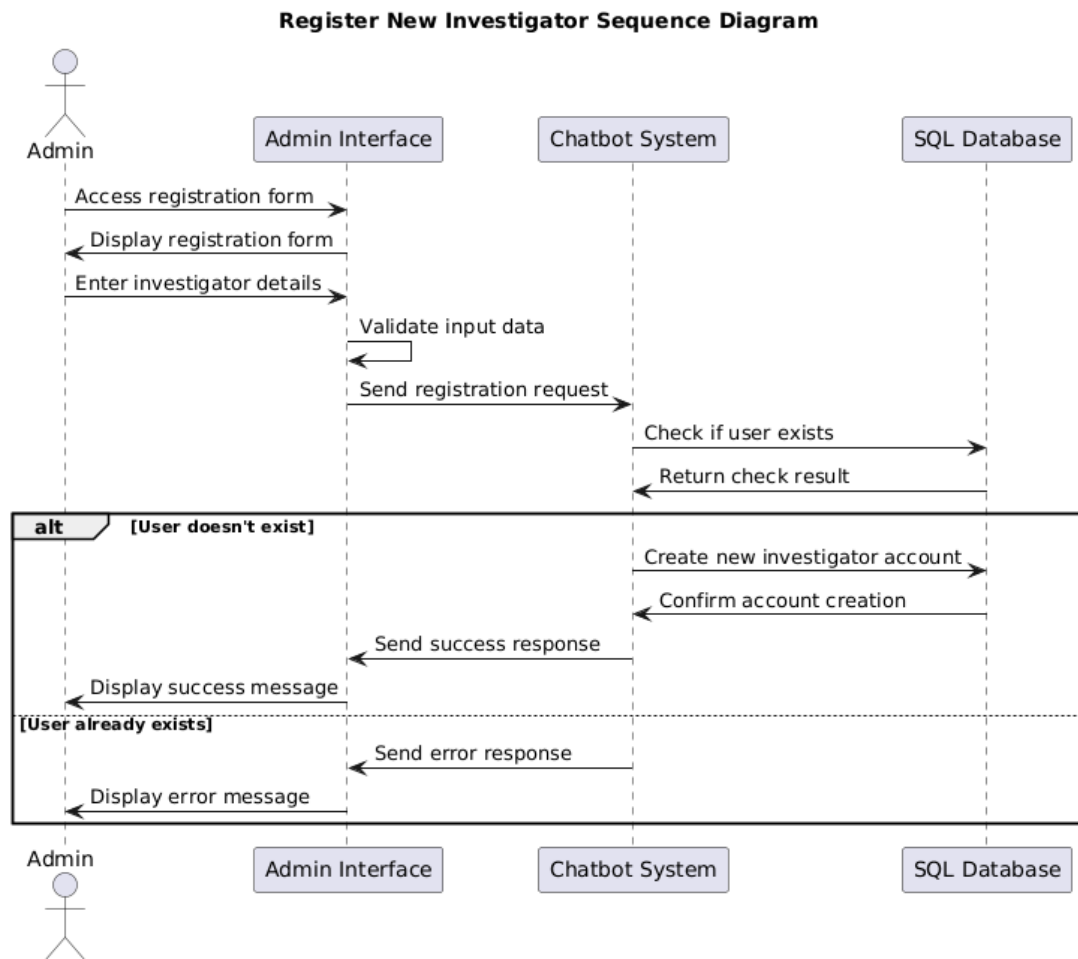


Figure A.12 Register New Investigator Sequence Diagram

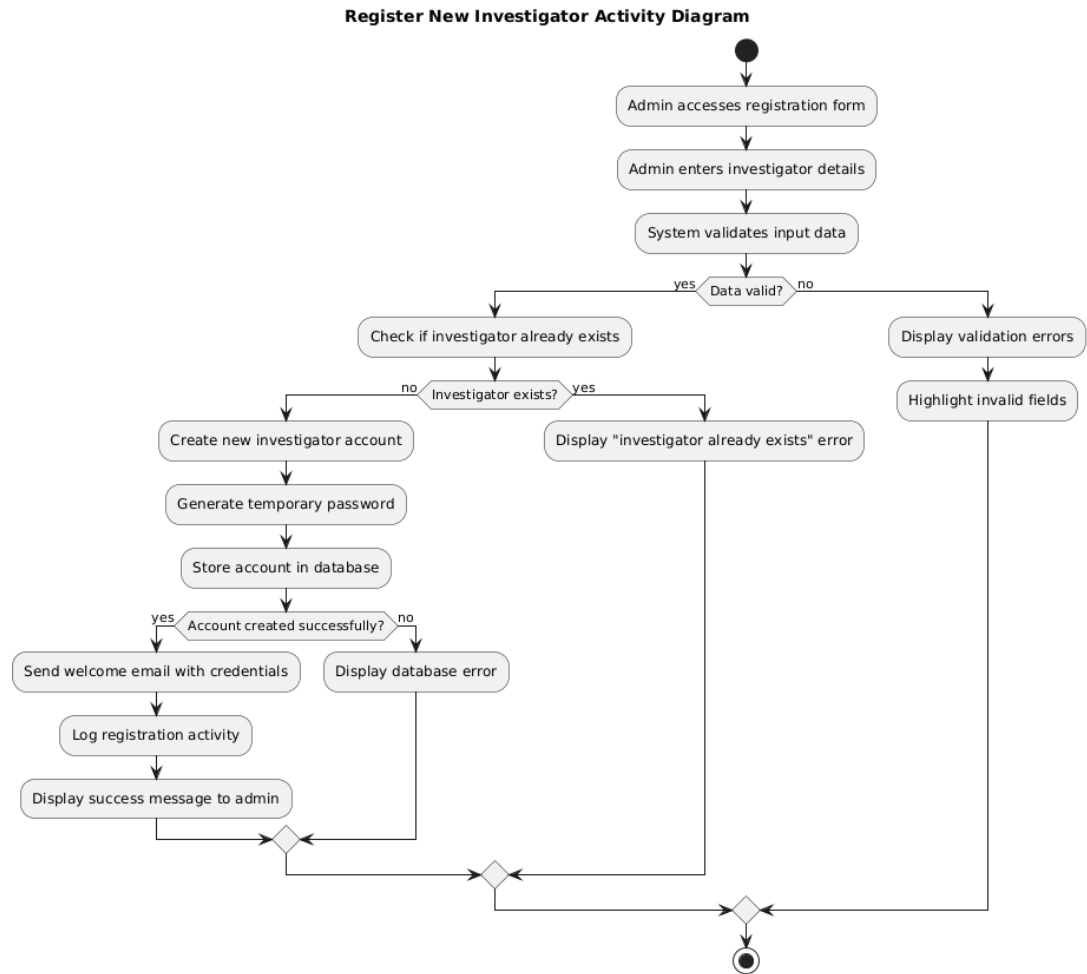


Figure A.13 Register New Investigator Activity Diagram

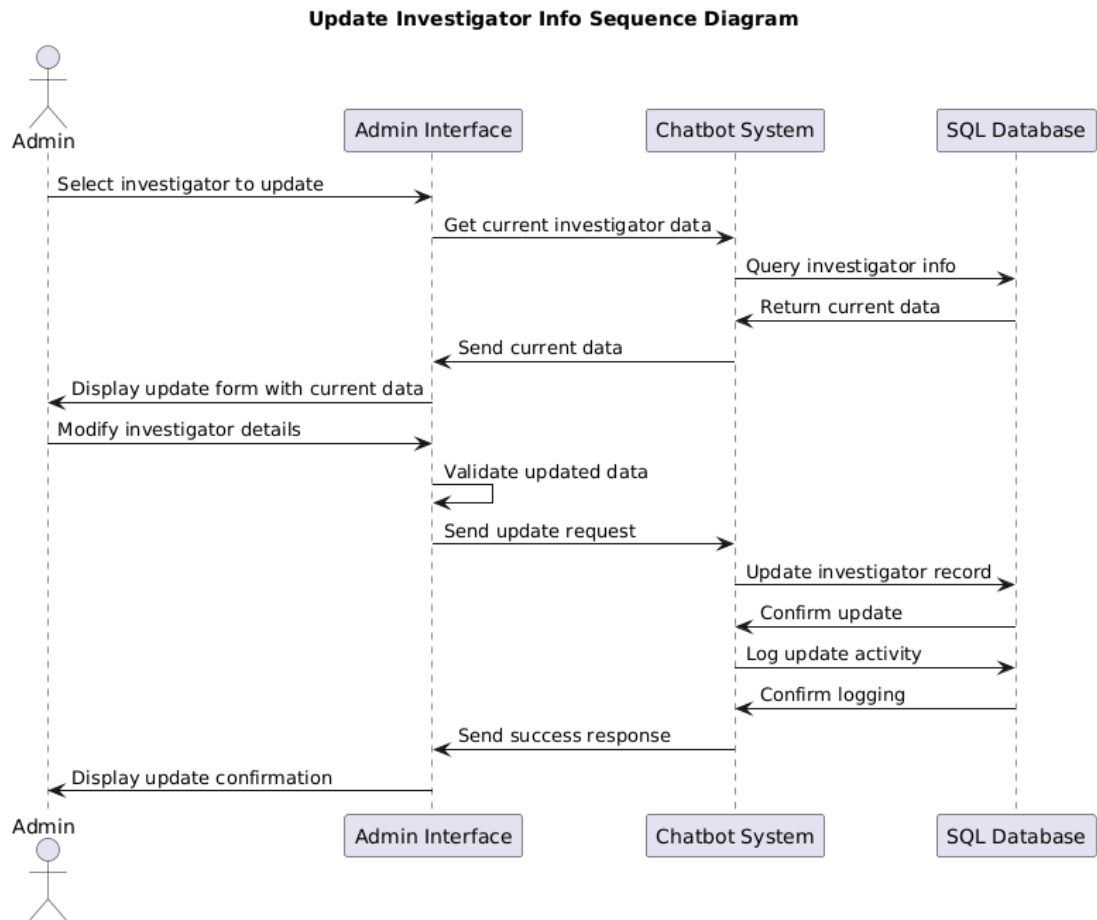


Figure A.14 Update Investigator Info Sequence Diagram

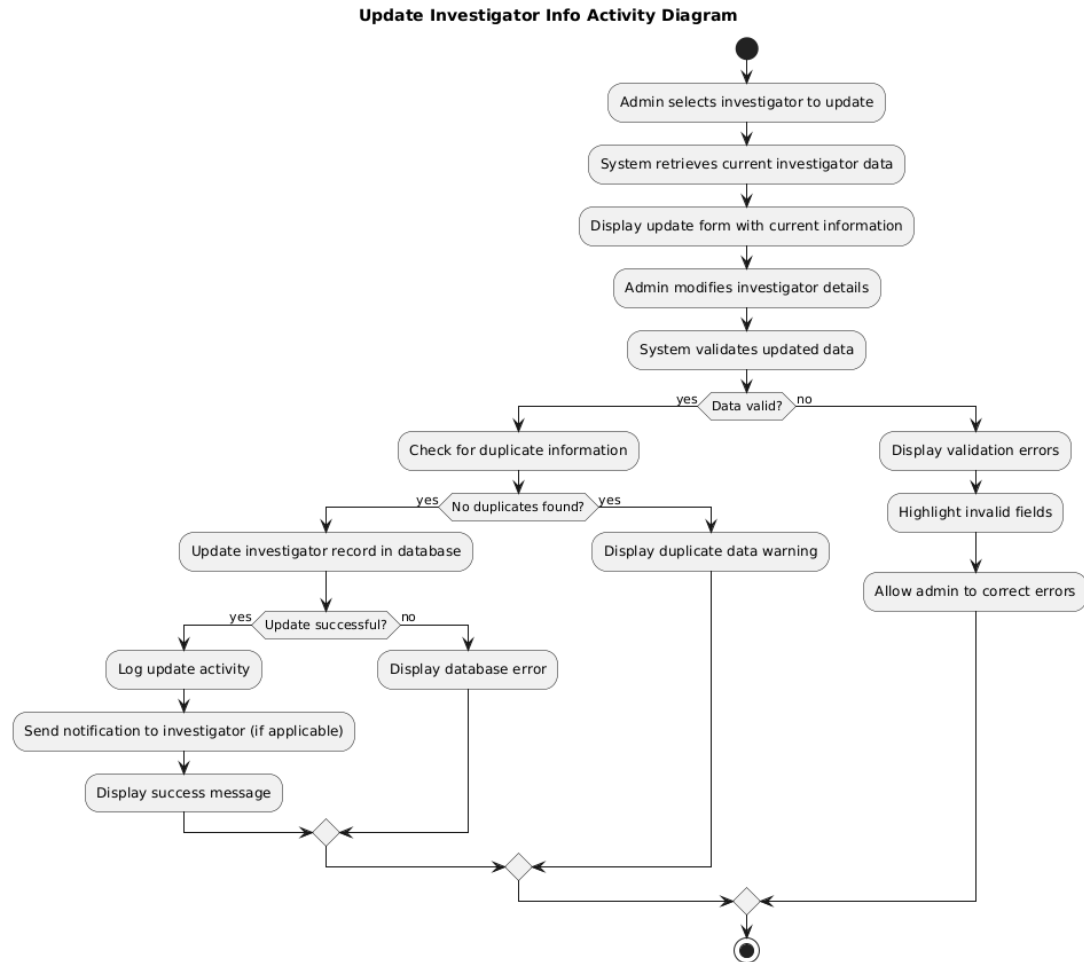


Figure A.15 Update Investigator Info Activity Diagram

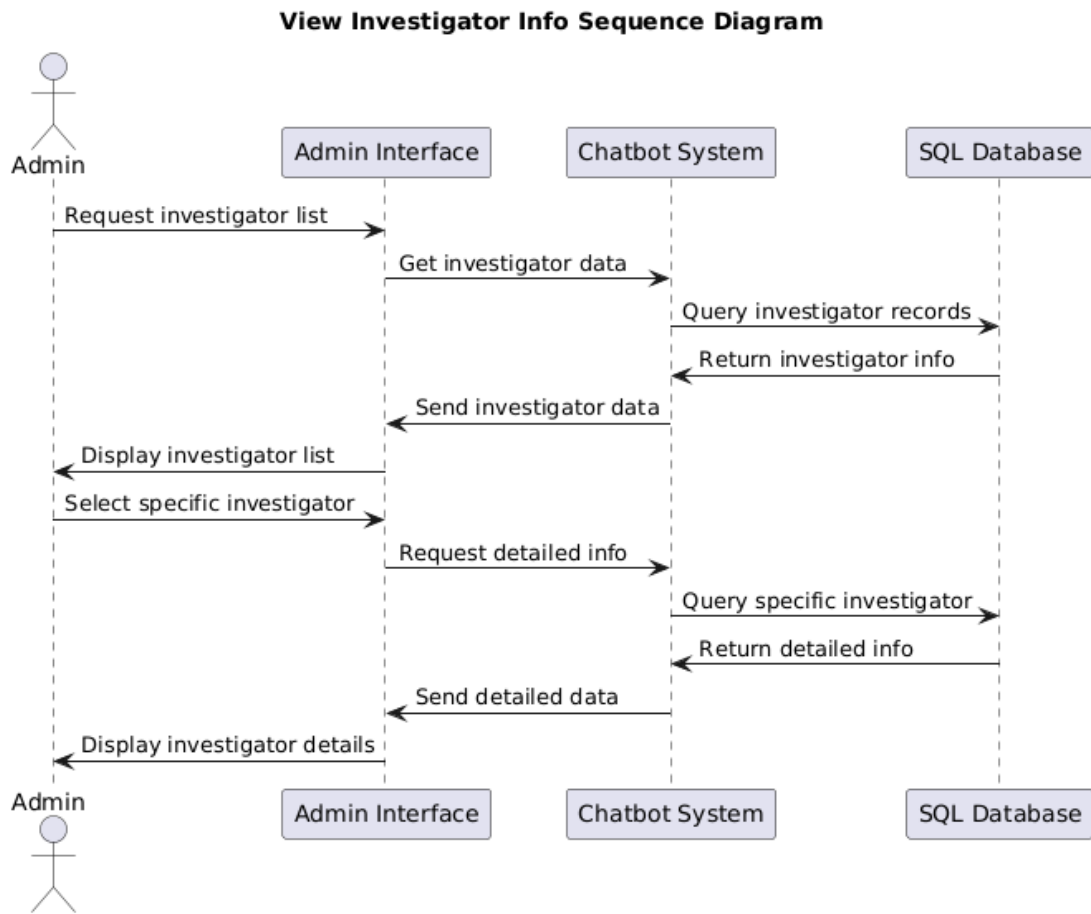


Figure A.16 View Investigator Info Sequence Diagram

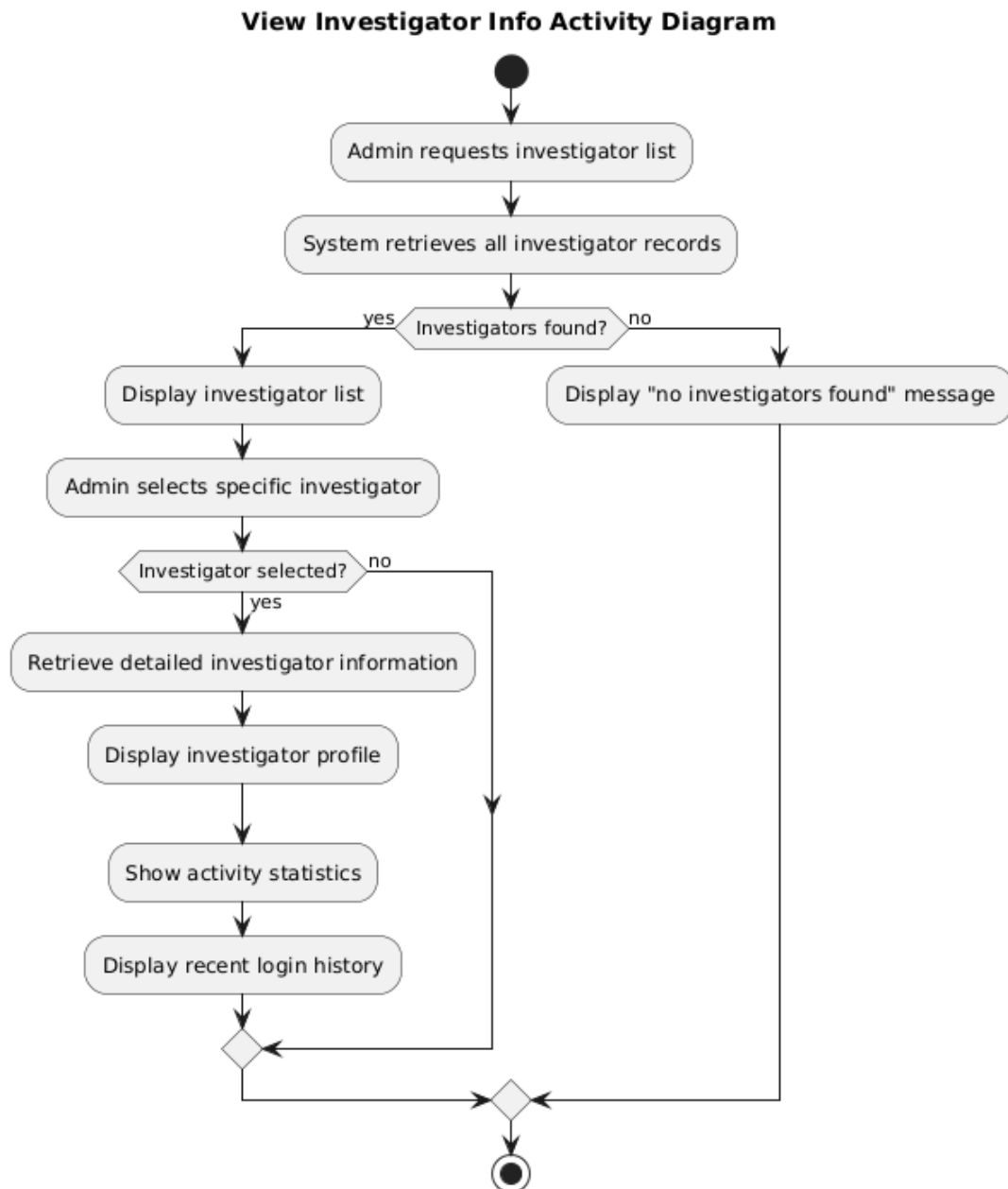


Figure A.17 View Investigator Info Activity Diagram

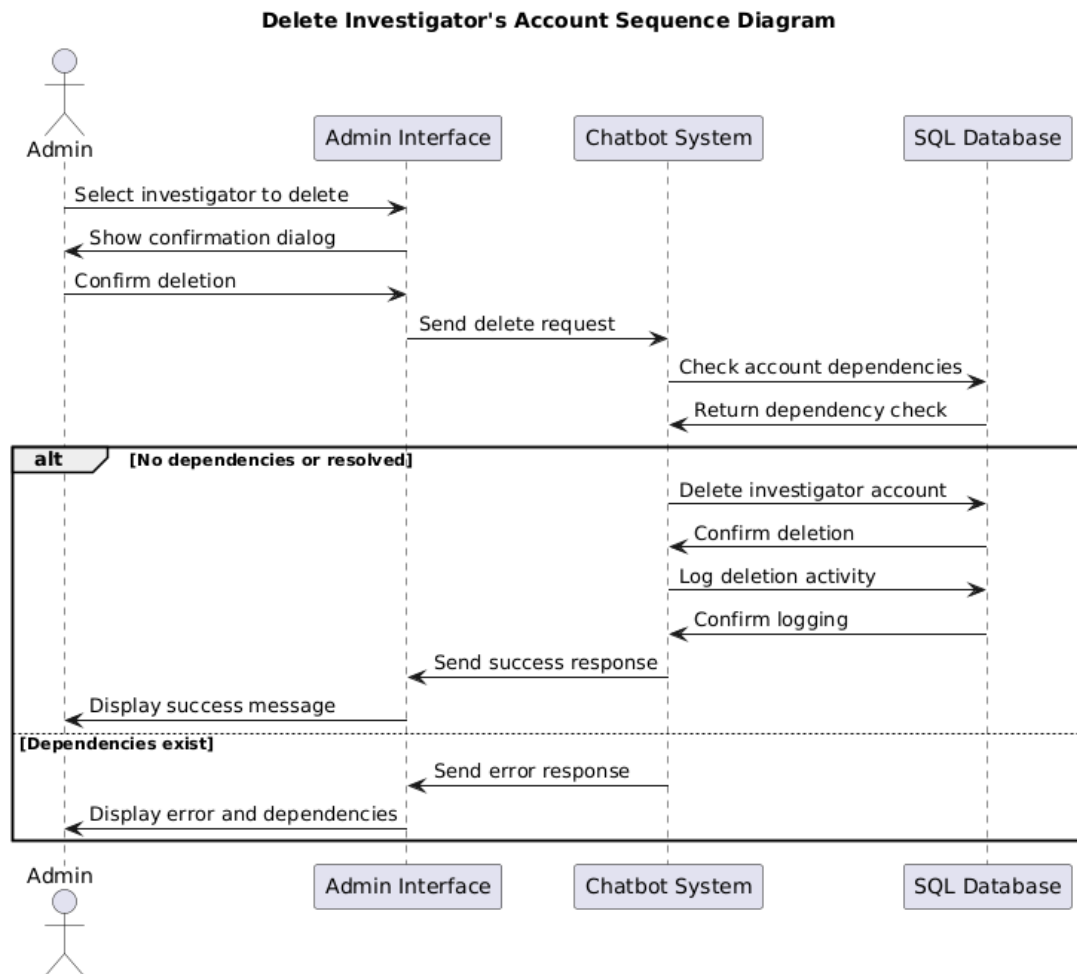


Figure A.18 Delete Investigator's Account Sequence Diagram

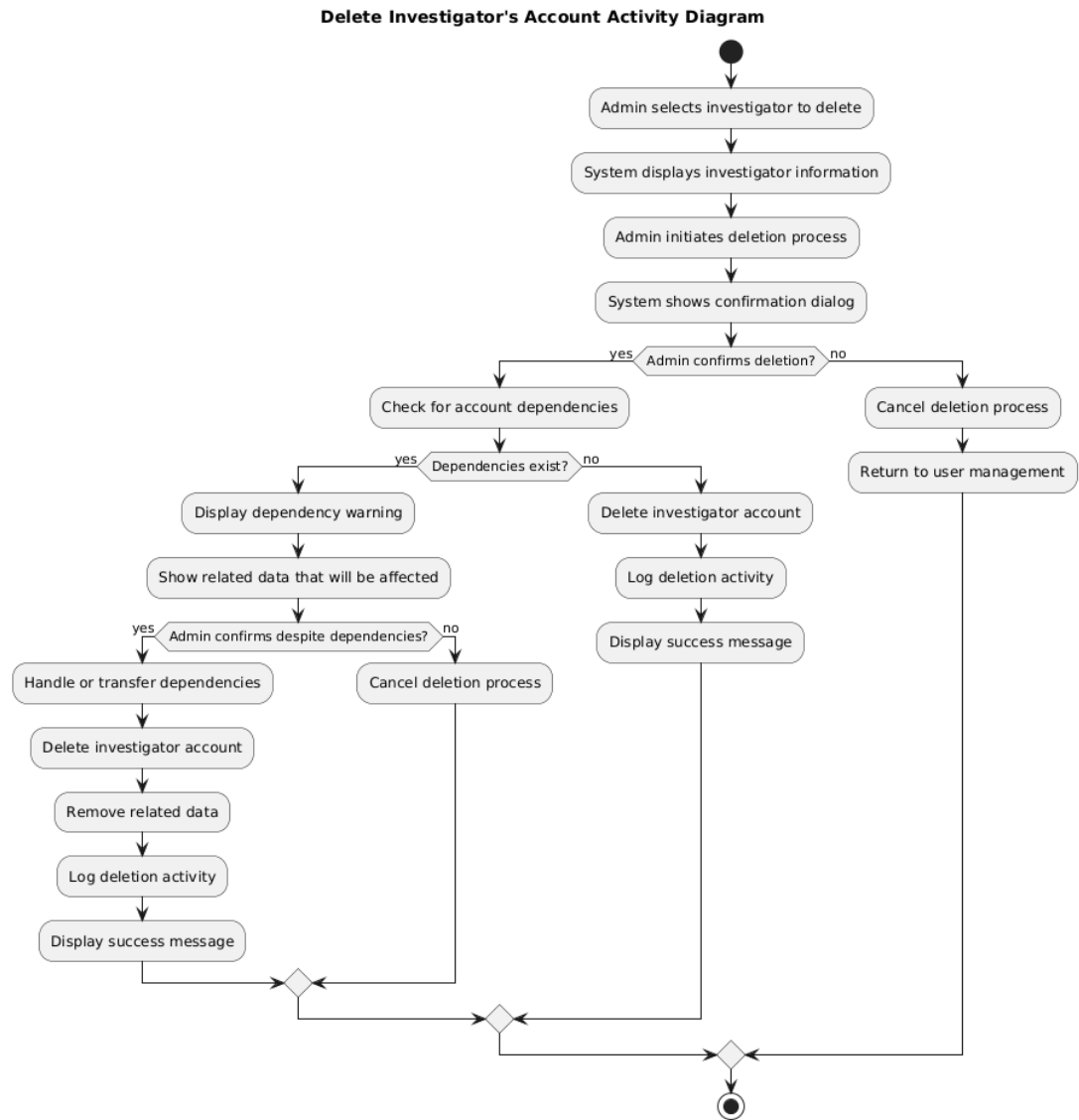


Figure A.19 Delete Investigator's Account Activity Diagram

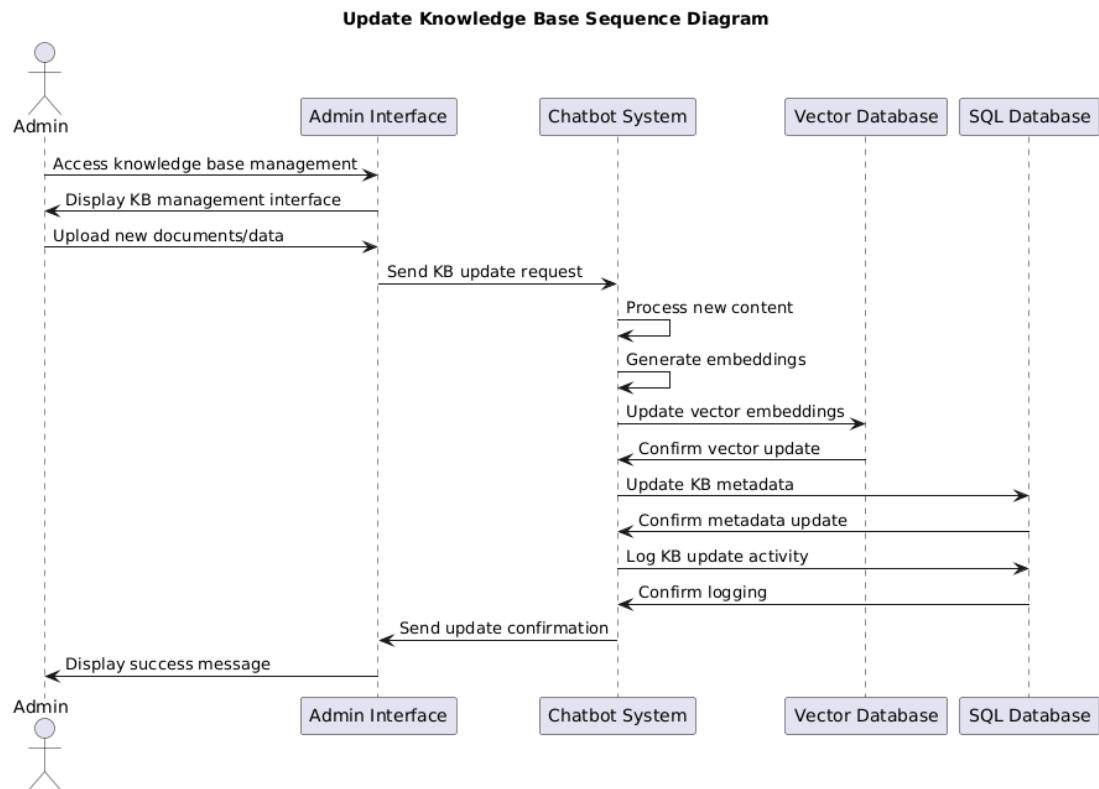


Figure A.20 Update Knowledge Base Sequence Diagram

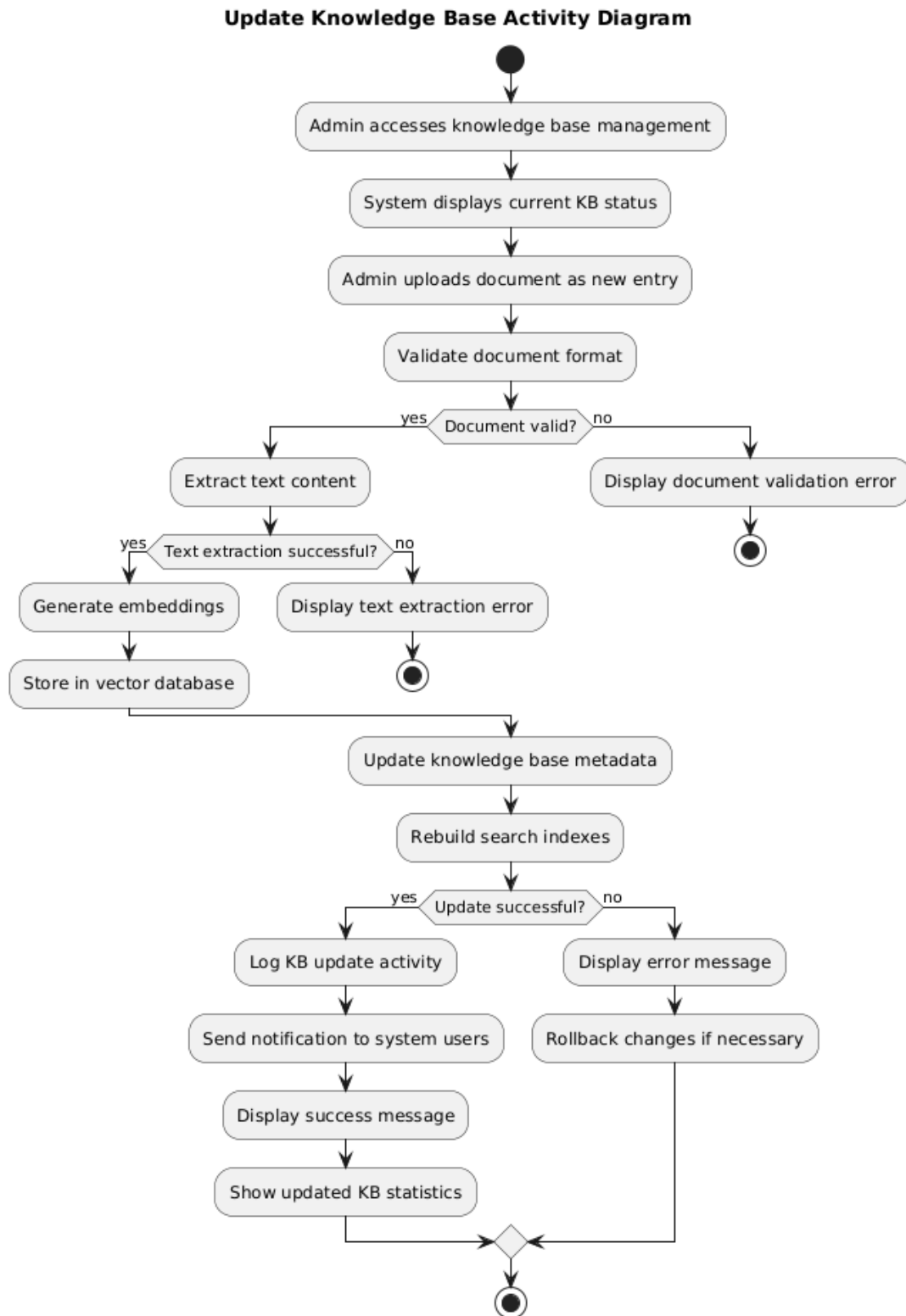


Figure A.21 Update Knowledge Base Activity Diagram

Appendix B PSM Gantt Chart

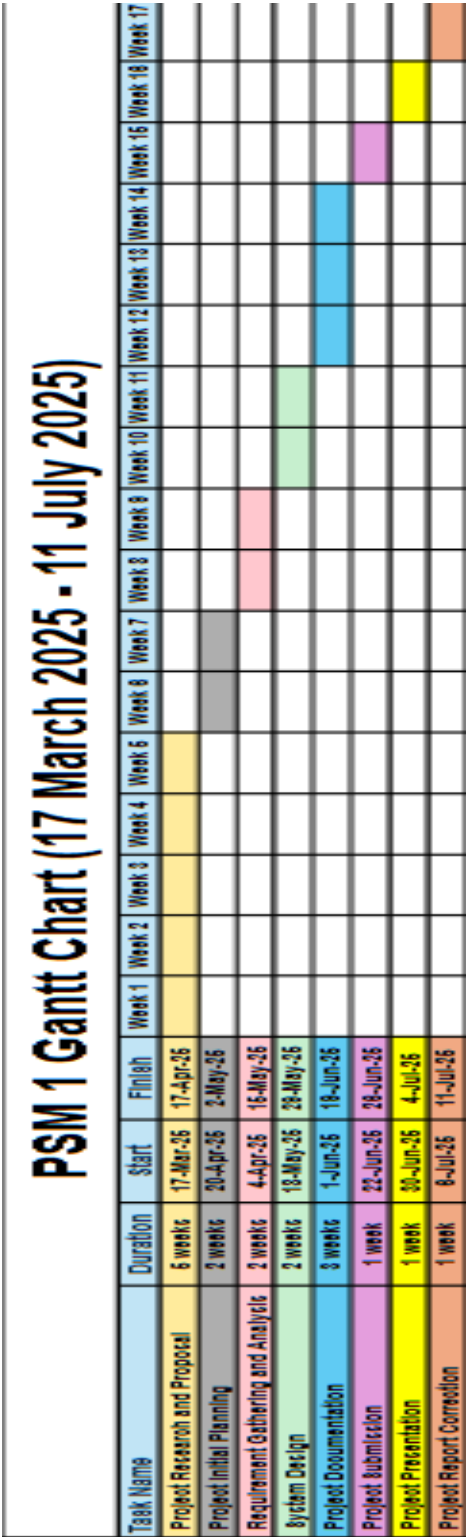


Figure B.1 PSM1 Gantt Chart

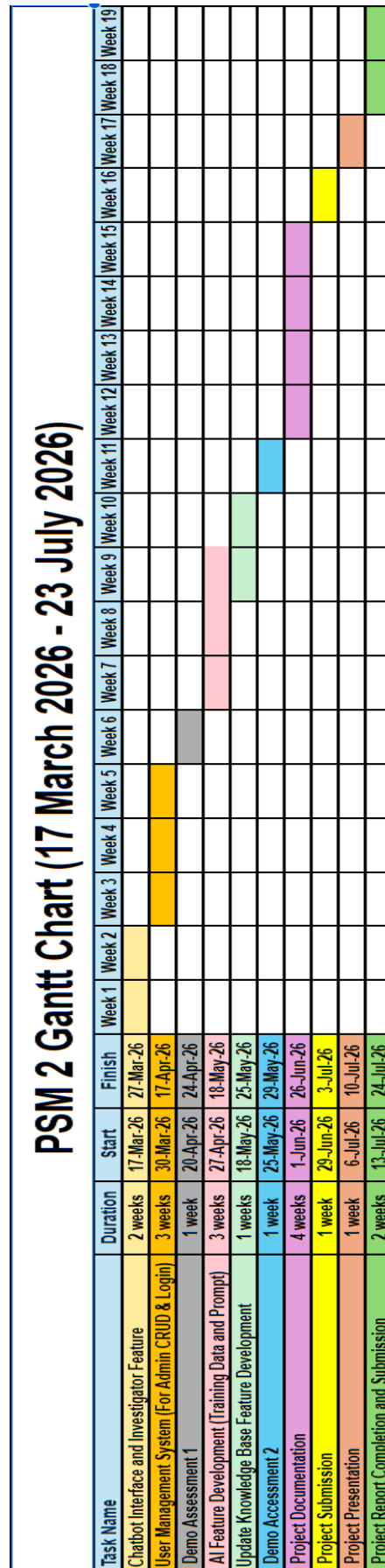


Figure B.2 PSM2 Gantt Chart

Appendix C Data Source

The screenshot shows the elaw.my website interface. The search bar at the top contains the text 'money laundering'. Below the search bar, there are tabs for 'Results', 'Advanced Search', and 'Case Citations'. The 'Results' tab is selected, showing a list of cases. The first case is 'HONG LEONG BANK BERHAD V. TEN JING ENTERPRISE SDN BHD & ORS AND ANOTHER APPEAL'. The second case is 'CHAN V. SHAN & ORS; ALIR (INTERVENER)'. The third case is 'PP V. MUHAMAD NORULAINI SUPARDI'. The fourth case is 'CIMB ISLAMIC BANK BERHAD V. SURUHANJAYA PENCEGAHAN RASUAH MALAYSIA & ORS'. The fifth case is 'ANISH RESOURCES SDN BHD V. PUBLIC BANK BHD'. Each case entry includes the case name, the judge, a brief description, and the date of the judgment.

Figure C.1 Example 2 of Financial Crime Cases from elaw.my

The screenshot shows the elaw.my website interface with a more specific search query: 'financial crime fraud money laundering'. The search bar at the top contains this text. Below the search bar, there are tabs for 'Results', 'Advanced Search', and 'Case Citations'. The 'Results' tab is selected, showing a list of cases. The first case is 'CHAN V. SHAN & ORS; ALIR (INTERVENER)'. The second case is 'ROSLI DAHLAN V. DATUK AHMAD TERRIRUDIN MOHD SALLEH & ORS'. The third case is 'ROSLI DAHLAN V. IG TAN SRI RAZARUDIN HUSAIN @ ABD RASHID & ORS'. The fourth case is 'DR ZAKIR ABDUL KARIM NAIK V. RAMASAMY PALANISAMY & ANOTHER CASE'. Each case entry includes the case name, the judge, a brief description, and the date of the judgment.

Figure C.2 Example 3 of Financial Crime Cases from elaw.my

Appendix D User Interfaces

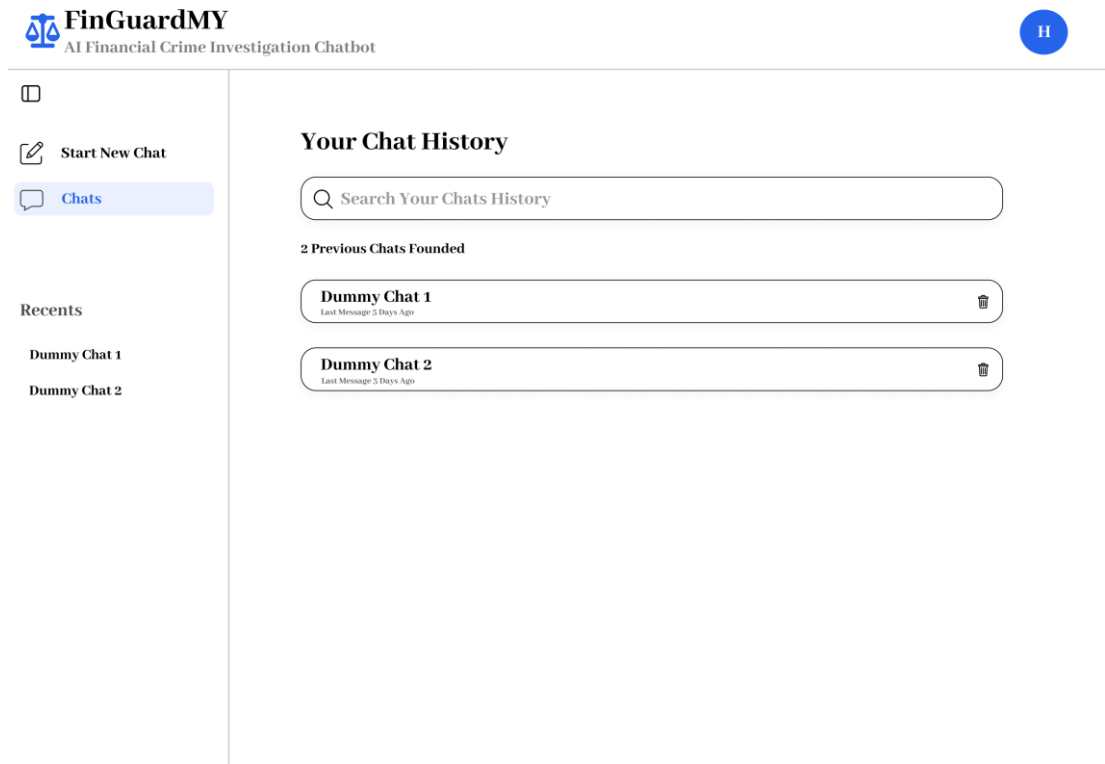


Figure D.1 View Chat History Interface

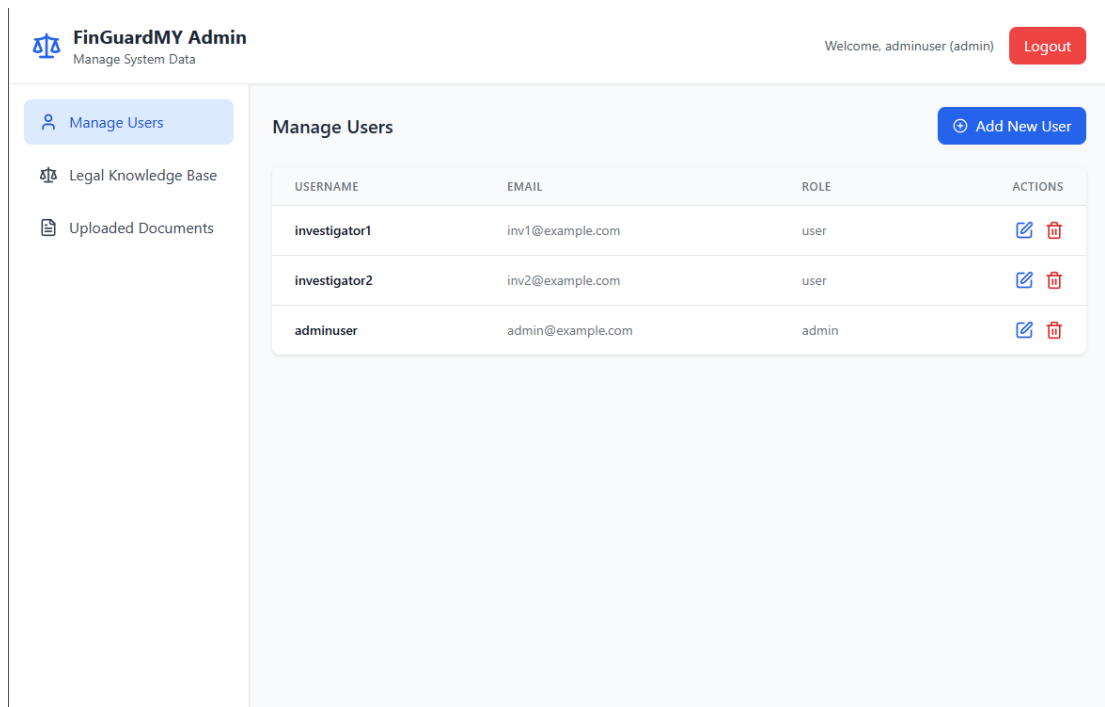


Figure D.2 Manage User Interface

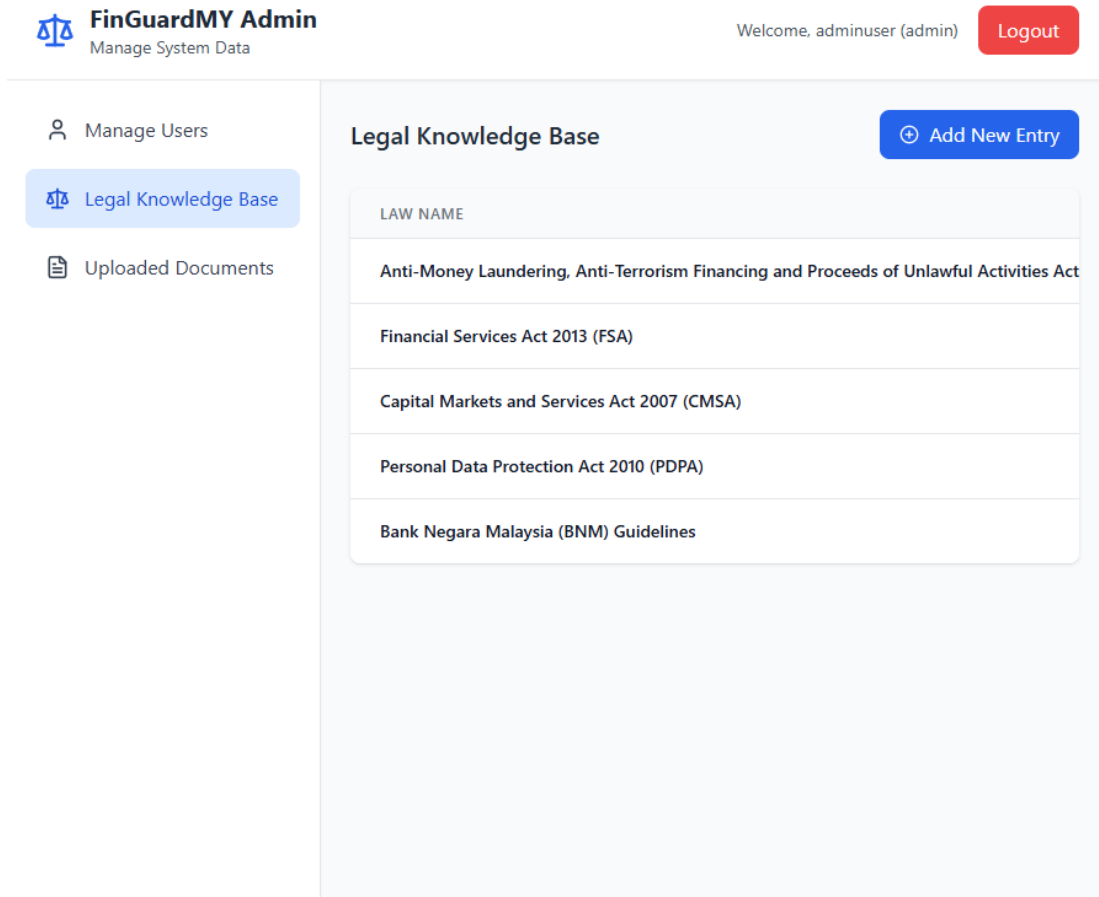


Figure D.3 Legal Knowledge Base Interface

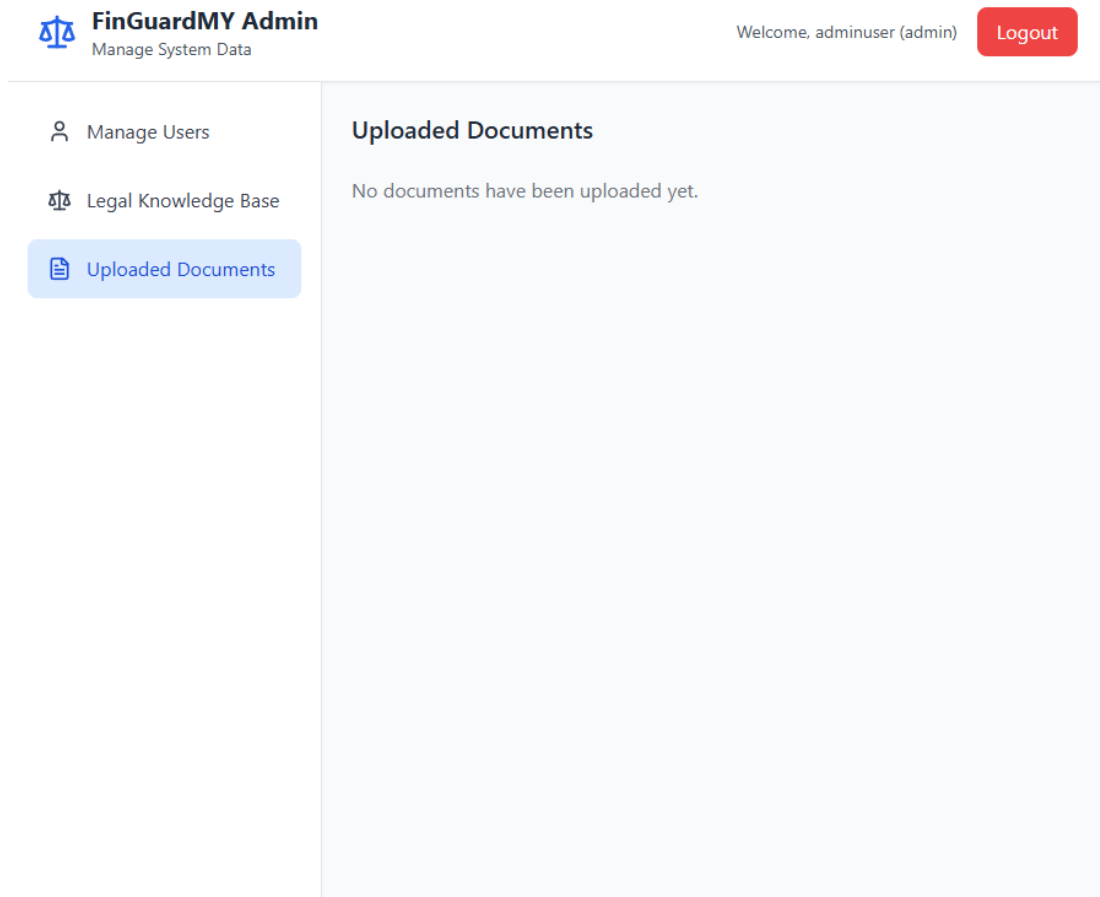


Figure D.4 View Uploaded Document Interface